

# Species distribution models can be highly sensitive to algorithm configuration

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## ARTICLE INFO

### Keywords:

Configuration option settings  
Provenance  
Transparency  
Koala  
Thorny devil  
MaxEnt  
ANN GLM  
GAM  
MARS  
FDA  
SRE  
CTA

## ABSTRACT

In pursuit of a more robust provenance in the field of species distribution modelling, an extensive literature search was undertaken to find the typical default values, and the range of values, for configuration settings of a large number of the most commonly used statistical algorithms available for constructing species distribution models (SDM), as implemented in the R script packages (such as Dismo and Biomod2) or other species distribution modelling programs like MaxEnt. We found that documentation of SDM algorithm configuration option settings in the SDM literature is, overall, very uncommon, and the justifications for these settings were minimal, when present. Such settings were often the R default values, or were the result of trial and error. This is potentially concerning since: (i) it detracts from the robustness of the provenance for such SDM studies; (ii) a lack of documentation of configuration option settings in a paper prevents the replication of an experiment, which contravenes one of the main tenets of the scientific method; (iii) inappropriate or uninformed configuration option settings are particularly concerning if they represent a poorly understood ecological variable or process, and if the algorithm is sensitive to such settings, this could result in erroneous and/or unrealistic SDMs. Therefore, this study sets out to comprehensively test the sensitivity of eight widely used SDM algorithms to variation in configuration options settings: MaxEnt, Artificial Neural Network (ANN), Generalized Linear Model (GLM), Generalized Additive Model (GAM), Multivariate Adaptive Regression Splines (MARS), Flexible Discriminant Analysis (FDA), Surface Range Envelope (SRE) and Classification tree analysis (CTA). A process of expert elicitation was used to derive a range of appropriate values with which to test the sensitivity of our algorithms. We chose to use species occurrence records for two species - Koala (*Phascolarctos cinereus*) and Thorny Devil (*Moloch horridus*) - in order to investigate how algorithm sensitivity depends on the species being modelled. Results were assessed by comparing the modelled distribution of the control SDM (default settings) to the modelled distribution from each sensitivity test SDM (i.e. non-default configuration settings). This was done using the visual and statistical measures of predictive performance available in the Biodiversity and Climate Change Virtual Laboratory (BCCVL), including the area under the (receiver operating characteristic) curve. The aim of our study was to be able to draw conclusions as to how the sensitivity of SDM algorithms to their configuration option settings may detract from the reliability of SDM results, given the often unjustified and unscrutinized use of the default settings, and generally infrequent and largely perfunctory attendance to this issue in most of the published SDM literature. Our results indicate that all of the algorithms tested showed sensitivity to alternative (non-default) values for some of their configuration settings and that often this sensitivity is species-dependent. Therefore we can conclude that the choice of configuration settings in these widely used SDM algorithms can have a large impact on the resulting projected distribution. This has important ramifications for decision-making and policy outcomes wherever SDMs are used to inform species and biodiversity management plans and policy settings. This study demonstrates that assigning suitable values for these settings is a very important consideration and as such should always be published along with the model. Documenting all configuration settings is necessary to increase the scientific robustness, transparency and reproducibility of species distribution modelling studies.

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<https://doi.org/10.1016/j.ecolmodel.2019.108719>

Received 26 October 2018; Received in revised form 12 June 2019; Accepted 13 June 2019

Available online 19 July 2019

0304-3800/ © 2019 Published by Elsevier B.V.

## 1. Introduction

Progress made in computational science in recent years has produced concomitant gains and exciting developments in many scientific disciplines. Many researchers in the computational sciences have increasingly advocated for reproducibility of research, which involves the data and computer code used in a published study to be made available to others, as a minimum standard when assessing the validity of scientific claims. Cassey and Blackburn (2006) have argued that in order for a scientific study to be acceptable for publication, it should be reproducible (NRC, 2003). They contend that “reproducibility is increasingly being requested by journals”. This standard of reproducibility is based on the theoretically available detailed log of every action taken using computers, which underlies every computational experiment. If this code were available for scrutiny, such transparency in research would surpass the analogous non-computational experimental descriptions printed in journals using a human language (Peng, 2011). For scientific fields which inform policy and management decisions, and in which there is difficulty in reproducing field experiments and techniques due to the non-replicable nature of environmental conditions, there is a critical need to track the provenance of derived data products and scientific results. ‘Provenance’, which is defined as “information about entities, activities, and people involved in producing a piece of data or thing, can be used to form assessments about the quality, reliability or trustworthiness” (W3C Working Group, 2013) of scientific results. Provenance should cover initial data collection, quality assurance, analyses, modelling and publication (Reichman et al., 2011).

It was with a view towards more robust provenance in the field of species distribution modelling (SDM), that a literature search was undertaken to find the typical default values, and an appropriate range of values for configuration option settings in ten algorithms available for constructing species distribution models in the Biodiversity and Climate Change Virtual Laboratory (BCCVL) (Hallgren et al., 2016). The results of this literature search were summarized in Hallgren and Mackey (2014). The BCCVL ([www.bccvl.org.au](http://www.bccvl.org.au)) is an online, cloud-based virtual laboratory which brings together a multitude of datasets, SDM algorithms and several different modelling experiment types to create a highly accessible modelling platform which can be used to investigate the impact of climate change and other driving variables on species distributions, species traits, and several measures of biodiversity in Australia and around the world (Hallgren et al., 2016). The sources of the default values of packages in R (R Core Team, 2017), which provided the specific implementation of the algorithms used in the BCCVL, were also investigated. Hallgren and Mackey (2014) found that most SDM studies do not publish SDM configuration option settings, and if they do, then justifications for these parameter values, are minimal, and often rely on the R default values, or are the result of trial and error.

Consequently, we contend that it is advisable to document and publish the rationale for SDM configuration option settings for the SDM algorithms implemented in the R script packages, such as Dismo (Hijmans and Elith, 2013) and biomod2 (Thuiller et al., 2009) or other species distribution modelling programs, like MaxEnt (Phillips et al., 2006). If no information for SDM configuration option settings is provided in a publication, it makes the SDM experiments unable to be replicated, which contravenes one of the main tenets of the scientific method, and presents a barrier to good provenance.

Therefore, we argue that the explicit documentation of settings is critical for the provenance of a modelling study. It is also good scientific practice as it promotes methodological transparency and reproducibility, both of which are critical to progress in this field. Greater transparency of model configuration settings may be useful in identifying settings which might be inappropriate or unrealistic, which would be particularly concerning if they represent a poorly understood ecological or physical process, or if the algorithm was sensitive to such settings.

Thus, we also argue that it is important to know how sensitive the modelling algorithm is to its configuration settings. Without knowing how sensitive an algorithm is to its configuration settings, inappropriately assigned values, i.e. values without reasonable scientific justification, could possibly lead to erroneous and/or unrealistic SDM results. Furthermore, the use of inappropriate and/or uninformed configuration settings which the algorithm is sensitive to could compromise the rigour of the modelling process and the validity of the results.

Within the considerable body of SDM literature, there have only been a small number of studies which investigate the sensitivity of SDM algorithms, and usually to one aspect of the modelling process e.g.; sensitivity to pseudoabsence selection in seven different SDM (Barbet-Massin et al., 2012), or sensitivity to input data (Pirathiban et al., 2015). Merow et al. (2013) provides a user's guide to MaxEnt's model settings, (background data, features, regularization and sampling bias) and provides broad guidelines on how to set these at the start of the modelling process. However, it doesn't go into specific detail about the sensitivity of MaxEnt to different values for these settings, although they recommend that would-be users “always explore how different setting choices affect predictions and report these; if predictions show substantial differences across settings, it is critical to have a strong justification for the specific setting choice” (Merow et al., 2013: 1067). Elith et al. (2008) used a large dataset to illustrate the effects of several settings of BRT related to model fitting. Elith et al. (2011) provide a valuable discussion on the mechanics of model fitting (e.g. feature selection, constraints and regularization) in MaxEnt, and some exploration of how differences in these settings (e.g. background points and feature types) can influence the results. However they don't explore the sensitivity of MaxEnt to different settings of these configuration options. Elith and Graham (2009) tested the effect of changing the parameter settings in Random Forest pertaining to how many variables are randomly selected for evaluation at each split of the tree and how many trees are allowed in the ensemble. They also briefly acknowledged the potential impact of changes in parameterization on performance in MaxEnt, BRT and GLM, but only provide information on an analysis of the sensitivity to different sampling and weighting of pseudoabsence data in GLM and BRT.

Anderson and Gonzalez (2011) tested the sensitivity of MaxEnt to geographic sampling bias, model complexity (by varying the linear and quadratic feature types) and level of regularization. Syfert et al. (2013) tested the sensitivity MaxEnt SDMs to the effects of sampling bias and model complexity via fitting a wide variety of environmental response curves (i.e. functions in MaxEnt). Convertino et al. (2011a, b) investigated the output variability related to species sample size, accuracy of sample, resolution of environmental factors, and set of predictors, while Convertino et al. (2014) applied a Morris method sensitivity analysis to MaxEnt to investigate the importance of a species' distributions' drivers in space and time, including ‘intrinsic and exogenous input factors’ such as parameter settings. Beaumont et al. (2016) tested a range of SDMs regarding the ‘sensitivity’ to projected climate change scenarios in terms of their likelihood to simulate extreme changes in projected future distribution. Jarnevicha et al. (2017) evaluated combinations of five different algorithms (simple and complex generalized linear models, multivariate adaptive regression splines, MaxEnt, boosted regression trees, and random forest) and background or pseudoabsence methods (random, minimum convex polygon, and continuous and binary kernel density estimator) to assess the sensitivity of model outcomes to choices made. Iturbide et al. (2018) tested the sensitivity of GLM, Random Forest and MARS to pseudo-absence sampling in terms of the uncertainty in future projections of their target species derived from ten pseudo-absence realizations.

However, to date, there has been no systematic investigation of sensitivity to a wide range of configuration option settings for a wide range of SDM algorithms. This study reports on research designed to address this gap, and expands upon the work started by Hallgren et al.

**Table 1**

The algorithms tested for sensitivity to input configuration options, the list of configurable options in the BCCVL, and the settings for those options used in the Koala and Thorny Devil SDM sensitivity testing, as well as the default setting as implemented in the BCCVL.

| Algorithm                                       | Configurable options*                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Default                                                                     | Settings tested                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum Entropy model (MaxEnt)                  | 1) Weighted response weights<br>2) Maximum number of iterations<br>3) Linear/quadratic/product/ threshold/hinge feature thresholds<br><br>4) Product Feature thresholds<br><br>5) Quadratic feature threshold<br>6) Hinge feature threshold<br>7-11) Regularisation options:<br>- threshold feature regularization<br>- categorical feature regularization<br>- linear/quadratic/product feature regularization'<br>- hinge feature regularization<br>- Regularization multiplier<br>12) Prevalence | 0.5<br>200<br>Yes<br><br>80<br><br>10<br>15<br>1<br><br><br><br><br><br>0.5 | 0.25, 0.33, 0.66, 0.75, 0.9<br>2, 5, 10, 100<br>No (linear)<br>No (linear/quadratic)<br>No (Linear/quadratic/product)<br>No (Linear/quadratic/product/ threshold)<br>No (Linear/quadratic/product/ threshold/hinge)<br>40 (the Product Feature threshold changes from 10 to 5, and hinge feature threshold from 15 to 8)<br>5<br>8<br>2, 5, 10<br><br><br><br><br><br>0.3, 0.25, 0.1 |
| Artificial Neural Network (ANN)                 | 1) Weighted response weights<br>2) NbCV                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Null<br>5                                                                   | 0.66, 0.75, 0.9, 0.25, 0.33<br>10, 20                                                                                                                                                                                                                                                                                                                                                |
| Generalised Linear Model (GLM)                  | 1) Weighted response weights<br>2) Type<br>3) Interaction level                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0.5<br>quadratic<br>1                                                       | 0.66, 0.75, 0.9, 0.25, 0.33<br>polynomial<br>2                                                                                                                                                                                                                                                                                                                                       |
| Generalized Additive Model (GAM)                | 4) Test<br>1) Weighted response weights<br>2) Interaction level<br>3) irls.reg<br>4) Epsilon<br>5) Max MLE iterations<br>6) Convergence tolerance<br>7) Number of halvings                                                                                                                                                                                                                                                                                                                          | AIC<br>0.5<br>0<br>0.0<br>0.000001<br>100<br>1 E-7<br>15                    | None, BIC<br>IF P:A Ratio > 1; 0.55, 0.6, 0.9, 0.99<br>1, 2<br>0.1, 0.5, 0.9,<br>0.000005, 0.00001, 0.00005, 0.0001, 0.0005<br>50, 150, 200,<br>1 E-6, 1 E-5<br>5, 10, 20, 30                                                                                                                                                                                                        |
| Multivariate Adaptive Regression Splines (MARS) | 1) Degree<br>2) NK (max. no. of terms)<br>3) Penalty<br><br>4) Thresh<br>5) Prune<br>6) Weighted response weights<br>7) Resampling                                                                                                                                                                                                                                                                                                                                                                  | 1<br>Null<br>2<br><br>0.001<br>True<br>0.5<br>0                             | 2<br>5, 10,15, 20,100<br>If 'degree' = 1: 2, 2.5<br>If 'degree' = 2: 3, 4<br>0.05, 0.01, 0.005, 0.001<br>False<br>IF P:A RATIO > 1; 0.55, 0.6, 0.9, 0.99<br>0, 10, 20, 100                                                                                                                                                                                                           |
| Flexible Discriminant Analysis (FDA)            | 1) Weighted response weights<br>2) Resampling<br>3) Method                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.5<br>0<br>mars                                                            | IF P:A RATIO > 1; 0.55, 0.6, 0.9, 0.99<br>0, 10, 20, 100<br>polyreg, bruto, gen.ridge                                                                                                                                                                                                                                                                                                |
| Surface Range Envelope (SRE)                    | 1) Weighted response weights<br>2) Resampling<br>3) Quantile                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0.5<br>0<br>0.025                                                           | IF P:A RATIO** > 1; 0.55, 0.6, 0.9, 0.99<br>0, 10, 20, 100<br>0.01, 0.025, 0.05, 0.1                                                                                                                                                                                                                                                                                                 |
| Classification Tree Analysis (CTA)              | 1) Weighted response weights<br>2) Cross-validation folds<br>3) Minimum bucket<br>4) Minimum split<br>5) Complexity parameter<br>6) Maximum depth                                                                                                                                                                                                                                                                                                                                                   | 0.5<br>10<br>1<br>20<br>0.01<br>30                                          | 0.25, 0.33 0.66, 0.75, 0.9<br>5, 10, 20<br>50, 100, 1000<br>50, 100, 500, 1000<br>0.025, 0.05<br>10, 20, 40                                                                                                                                                                                                                                                                          |

\* All configuration options included in sensitivity tests are described in Appendix 1.

\*\* Presence-Absence ratio.

(2017), which focused on analyzing the sensitivity of Boosted Regression Trees and Random Forests to configuration settings, to eight more species distribution modelling algorithms. This study aims to: (1) comprehensively test the sensitivity of eight algorithms which are commonly used in species distribution modelling to variation in their configuration option settings; (2) draw conclusions as to how the sensitivity of SDM algorithms may impact on the reliability of SDM results, given the frequently unjustified nature of the default algorithm settings, and the generally low level of awareness of, and attendance to, this issue in much of the published literature (keeping in mind that this reliability is determined by a number of factors including the algorithm chosen, the available occurrence data and the objective of the

modelling study); (3) articulate why setting reasonable and justifiable settings is important for the algorithms tested, and which configuration options it is most important to be careful setting, given the algorithm's sensitivity to them.

## 2. Methodology

### 2.1. Data used

Occurrence (presence) datasets for the Koala (*Phascolarctos cinereus*) and Thorny Devil (*Moloch horridus*) were downloaded from the ALA and cleaned; the Koala dataset used in this study consists of 80,938 records,

and the Thorny Devil dataset has 759 records. As for Hallgren et al. (2017), we chose the Koala to test algorithm sensitivity to configuration settings due to its well-known distribution. In this study, we chose the Thorny Devil as a second species to test the algorithm's sensitivity to configuration settings, in order to understand how dependent algorithm sensitivity is to the species studied. We specifically chose the Thorny Devil as it has a completely different (and non-overlapping) observed distribution to the Koala, is far less prevalent, and it is in an entirely different class of animal (Reptilia as opposed to Mammalia). Habitat predictor variables were chosen as input into the species distribution models, according to the method used by Hallgren et al. (2017) for the Koala, so as to be able to compare our results to this previous study. These predictor variables comprised the WorldClim current climate (Hijmans et al., 2005) bioclimatic variables #5 (Max Temperature of Warmest Month), #9 (Mean Temperature of Driest Quarter), #10 (Mean Temperature of Warmest Quarter), #12 (Annual Precipitation), #14 (Precipitation of Driest Month), and #17 (Precipitation of Driest Quarter). All of these were available in the BCCVL.

For the Thorny Devil, after initially modelling its distribution using all 19 WorldClim variables, we chose predictor variables based on the variable importance plots and response curves derived from the BCCVL SDM output for this species; this was done due to the paucity of literature on ecophysiology and climatic drivers specific to the Thorny Devil. This approach is similar to that of Phillips et al. (2006) who advocate "including all reasonable predictors in the model (MaxEnt) and letting the algorithm decide which ones are important" (Merow et al., 2013). Twelve variables were found to be important across the range of algorithms, and hence used in our sensitivity tests; these were the WorldClim bioclimatic variables #1 (Annual mean temperature), #3 (Isothermality), #4 Temperature seasonality), #9 (Mean temperature of the driest quarter), #10 (Mean temperature of the warmest quarter), #11 (Mean temperature of the coldest quarter), #12 (Annual precipitation), #13 (Precipitation of the wettest month), #14 (Precipitation of the driest month), #16 (Precipitation of the wettest quarter), #17 (Precipitation of the driest quarter), #18 (Precipitation of the warmest quarter) and #19 (Precipitation of the coldest quarter).

## 2.2. Algorithms tested

Since different statistical models, of vastly different forms, may fit the same process equally well, with each providing a different perspective or "angle" onto the same phenomenon, it is not possible to know ahead of time, for a particular species, which "shape" or template for a SDM will fit well. For this reason, we control for this uncertainty by investigating more than one SDM algorithm for our modelled species.

Hallgren et al. (2017) examined two machine-learning algorithms for SDM (Franklin, 2010), which are both extensions of Classification Trees (CT: Breiman et al., 1984): Boosted Regression Trees (BRT: Breiman, 2001) and Random Forests (RF: Breiman, 1999, 2001). This study examines eight more algorithms commonly used for species distribution models, all of which are implemented in the BCCVL, as listed in Table 1: MaxEnt (Phillips et al., 2006; Phillips and Dudík, 2008), Artificial Neural Networks (ANN) (Hopfield, 1982; Lek and Guégan, 1999), Generalized Linear Model (GLM) (McCullagh and Nelder, 1989; Guisan et al., 2002), Generalized Additive Model (GAM) (Guisan et al., 2002), Flexible Discriminant Analysis (FDA) (Hastie et al., 1994; Trevor et al., 1994), Surface Range Envelope (SRE) (Busby, 1991; Jiguet et al., 2011), Multivariate Adaptive Regression Splines (MARS) (Friedman, 1991) and Classification tree analysis (CTA) (Breiman, 1984). Along with the two algorithms investigated in Hallgren et al. (2017), these eight algorithms were found to be widely used in the SDM literature.

*MaxEnt (Maximum Entropy modelling)* is a machine learning algorithm that uses presence data together with background locations. MaxEnt predicts species occurrences by finding the environmental profiles that lead to predictions that best differentiates presence from background, while taking into account the limits of the environmental variables of known locations. MaxEnt was tested for sensitivity to 12 configurable options in the BCCVL.

*Artificial Neural Networks* refers to a large group of models that are inspired by biological neural networks, in particular the brain, which consists of extremely large interconnected networks of neurons to process information. Artificial Neural Networks consist of a large number of nodes and connections which are typically organised in layers, with an input layer in which the data is fed into the model, a number of hidden layers, and the output layer which represents the result of the model. The most commonly used ANNs in ecology are single hidden layer feed-forward Artificial Neural Networks that are trained by back-propagation. ANN, GLM, GAM and MARS use both presence and absence (or pseudoabsence) data, as well as environmental data, and can handle both continuous and categorical predictors. ANN, as implemented in the BCCVL, was tested for sensitivity to 2 configurable options.

*Generalized Linear Models (GLM)* are an extension of 'simple' linear regression models, which predict the response variable as a weighted linear function of multiple predictor variables. Unlike simple linear regression, GLMs are able to deal with binary data such as presence/absence data. As implemented in the BCCVL, GLMs find the weights that make the species absence/presence data most likely, across the range of the environmental variables (maximum likelihood). In the BCCVL, linear or quadratic effects are available. GLM was tested for sensitivity to 4 configurable options.

*Generalized Additive Models (GAMs)* are an extension of GLMs in such a way that predictor variables can be modeled non-parametrically instead of being restricted like GLM to solely linear and polynomial terms for predictors. Therefore, GAMs tend to provide a close fit to the data, which can make them less transferrable, but is useful when the relationship between the variables are expected to be of a more complex form, not easily fitted by standard SDMs, or where there is no a priori reason for using a particular model. GAM was tested for sensitivity to 7 configurable options.

*Multivariate Adaptive Regression Splines (MARS)* is a non-parametric regression method that builds a decision tree with each branch comprising a linear regression model. It does this by partitioning the data, and runs a linear regression model on each different partition. The MARS algorithm is an extension of linear models that makes no assumptions about the form of the relationship between the response variable and the predictor variables. While GLMs and GAMs assume that the coefficients of the predictor variables are constant across all values of a predictor, the MARS algorithm specifically takes into account that this is often not the case. But the MARS algorithm has similarities to machine learning models such as tree-based models, because it achieves good fit by tuning predictive performance, via cross-validation. MARS was tested for sensitivity to 7 configurable options.

*Flexible Discriminant Analysis (FDA)* is a classification model based on a mixture of linear regression models, which uses optimal scoring to transform the weights in the regression for better linear separation, and multiple adaptive regression splines to generate the discriminant surface. FDA employs both presence and absence (or pseudoabsence) data.

*Surface Range Envelope (SRE)* is an envelope-style method similar to Bioclim (Nix, 1986). It uses presence-only data and the environmental conditions of locations of occurrence data to profile the environments where a species have been found. The envelope is defined by the minimum and maximum values of the environmental variables for all



**Table 2**

Evaluation statistics used to diagnose algorithm sensitivity to configuration settings in the Koala sensitivity analysis: (i) Area Under the Receiver Operating Curve (AUC), (ii) Positive Predictive Value (PPV) and (iii) Accuracy, for each of the control (default settings) and test species distribution models (with alternative configuration settings). Only the results for test settings to which the algorithms showed sensitivity are shown here. Bolded numbers indicate larger sensitivity.

| Algorithm                                                                                    | Configurable options               | Control AUC | Sensitive settings                              | Altered AUC | Control PPV | Sensitive settings | Altered PPV | Control Accuracy | Sensitive settings    | Altered Accuracy |      |
|----------------------------------------------------------------------------------------------|------------------------------------|-------------|-------------------------------------------------|-------------|-------------|--------------------|-------------|------------------|-----------------------|------------------|------|
| Maximum Entropy model (MaxEnt)                                                               | Weighted response weights          | 0.94        | 0.75                                            | 0.95        | 0.96        | 0.25               | 0.95        | 0.89             | 0.25, 0.66, 0.75, 0.9 | 0.91             |      |
|                                                                                              |                                    |             |                                                 |             |             | 0.33, 0.75, 0.9    | 0.95        |                  |                       |                  |      |
|                                                                                              |                                    |             |                                                 |             |             | 0.66               | 0.95        |                  |                       |                  |      |
|                                                                                              | Maximum number of iterations       |             | 10, 100                                         | 0.95        |             | 10, 100            | 0.95        |                  | 10                    | 0.90             |      |
|                                                                                              |                                    |             |                                                 |             |             |                    |             |                  | 100                   | 0.91             |      |
|                                                                                              |                                    |             | Threshold feature regularization                | 5, 10       |             | 0.95               | 2           |                  | 0.95                  | 2                | 0.91 |
|                                                                                              |                                    |             |                                                 |             |             | 10                 | 0.95        |                  | 5                     | 0.90             |      |
|                                                                                              |                                    |             |                                                 |             |             |                    |             |                  | 10                    | 0.92             |      |
|                                                                                              | Categorical feature regularization |             | –                                               | –           |             | 2                  | 0.95        |                  | 2                     | 0.91             |      |
|                                                                                              |                                    |             | Linear/quadratic/product feature regularization | 2, 10       |             | 0.95               | 5           |                  | 0.95                  | 5                | 0.90 |
|                                                                                              |                                    |             |                                                 |             |             |                    | 2           |                  | 0.95                  | 2                | 0.91 |
|                                                                                              |                                    |             |                                                 |             |             | 5                  | 0.95        |                  | 5, 10                 | 0.92             |      |
|                                                                                              | Hinge feature regularization       |             | 5                                               | 0.95        |             | 10                 | 0.95        |                  | 2                     | 0.88             |      |
|                                                                                              |                                    |             |                                                 |             |             |                    |             |                  | 5, 10                 | 0.91             |      |
|                                                                                              |                                    |             |                                                 |             |             |                    |             |                  |                       |                  |      |
|                                                                                              | Linear                             |             | –                                               | –           |             | –                  | –           |                  | ‘no’                  | 0.90             |      |
|                                                                                              | Linear&quadratic                   |             | –                                               | –           |             | –                  | –           |                  | –                     | –                |      |
|                                                                                              | Linear&quadratic&thres-hold        |             | –                                               | –           |             | –                  | –           |                  | –                     | –                |      |
| Product feature threshold = 40; quadratic feature threshold = 5; hinge feature threshold = 8 | PFT = 40                           | 0.95        | PFT = 40                                        | 0.95        | PFT = 40    | 0.92               |             |                  |                       |                  |      |
|                                                                                              | QFT = 5                            |             | QFT = 5                                         |             | QFT = 5     |                    |             |                  |                       |                  |      |
|                                                                                              | HFT = 8                            |             | HFT = 8                                         |             | HFT = 8     |                    |             |                  |                       |                  |      |
| Prevalence                                                                                   | 0.3                                | 0.95        | 0.1                                             | 0.95        | 0.1, 0.3    | 0.91               |             |                  |                       |                  |      |
|                                                                                              |                                    |             | 0.25                                            | 0.95        | 0.25        | 0.90               |             |                  |                       |                  |      |
|                                                                                              |                                    |             | 0.3                                             | 0.95        |             |                    |             |                  |                       |                  |      |
| Artificial Neural Network (ANN)                                                              | Weighted response weights          | 0.95        | 0.66                                            | 0.96        | 0.87        | 0.66               | 0.88        | 0.88             | 0.66, 0.90            | 0.90             |      |
|                                                                                              |                                    |             | 0.75                                            | 0.94        |             | 0.75               | 0.83        |                  |                       |                  |      |
|                                                                                              |                                    |             | 0.9                                             | 0.93        |             | 0.90               | 0.85        |                  |                       |                  |      |
| NbCV                                                                                         | 20                                 | 0.96        | 10                                              | 0.85        | 10          | 0.89               |             |                  |                       |                  |      |
|                                                                                              |                                    |             |                                                 |             | 20          | 0.90               |             |                  |                       |                  |      |
|                                                                                              |                                    |             |                                                 |             |             |                    |             |                  |                       |                  |      |
| Generalised Linear Model (GLM)                                                               | WRW Type                           | 0.94        | –                                               | –           | 0.84        | –                  | –           | 0.88             | –                     | –                |      |
|                                                                                              |                                    |             | polynomial                                      | 0.95        |             | polynomial         | 0.85        |                  | polynomial            | 0.89             |      |
|                                                                                              |                                    |             | 1                                               | 0.96        |             | 1                  | 0.87        |                  | 1                     | 0.90             |      |
| Interaction level                                                                            | 2                                  | 0.96        | 2                                               | 0.88        | 2           | 0.90               |             |                  |                       |                  |      |
|                                                                                              |                                    |             |                                                 |             |             |                    |             |                  |                       |                  |      |
|                                                                                              |                                    |             |                                                 |             |             |                    |             |                  |                       |                  |      |
| Generiszed Additive Model (GAM)                                                              | Test                               | 0.96        | –                                               | –           | 0.89        | –                  | –           | 0.90             | –                     | –                |      |
|                                                                                              |                                    |             | WRW                                             | –           |             | –                  | 0.60        |                  | 0.88                  | 0.99             | 0.89 |
|                                                                                              |                                    |             |                                                 |             |             |                    | 0.99        |                  | 0.88                  |                  |      |
| Interaction level                                                                            | –                                  | –           | –                                               | –           | –           | –                  | –           |                  |                       |                  |      |
|                                                                                              | Irls.reg                           | –           | –                                               | –           | –           | –                  | –           |                  |                       |                  |      |
|                                                                                              | Epsilon                            | –           | –                                               | –           | –           | –                  | –           |                  |                       |                  |      |
| Multivariate Adaptive Regression Splines (MARS)                                              | WRW                                | 0.96        | 0.90                                            | 0.94        | 0.87        | 0.55               | 0.88        | 0.90             | 0.90                  | 0.88             |      |
|                                                                                              |                                    |             | 0.99                                            | 0.93        |             | 0.90               | 0.84        |                  | 0.99                  | 0.87             |      |
|                                                                                              |                                    |             |                                                 |             |             | 0.99               | 0.82        |                  |                       |                  |      |
| Resampling                                                                                   | –                                  | –           | 100                                             | 0.88        | –           | –                  |             |                  |                       |                  |      |
|                                                                                              | NK (max. no. of terms)             | 5           | 0.93                                            | 5           | 0.83        | 5                  | 0.87        |                  |                       |                  |      |
|                                                                                              |                                    | 10          | 0.94                                            | 10          | 0.83        | 10                 | 0.87        |                  |                       |                  |      |
|                                                                                              |                                    |             | 15, 20                                          | 0.83        | 15          | 0.88               |             |                  |                       |                  |      |
|                                                                                              |                                    |             |                                                 |             | 20          | 0.88               |             |                  |                       |                  |      |
|                                                                                              |                                    |             |                                                 |             |             |                    |             |                  |                       |                  |      |
| Penalty Thresh                                                                               | –                                  | –           | –                                               | –           | –           | –                  |             |                  |                       |                  |      |
|                                                                                              | 0.05                               | 0.91        | 0.05                                            | 0.81        | 0.05        | 0.85               |             |                  |                       |                  |      |
|                                                                                              |                                    |             | 0.01                                            | 0.85        | 0.01        | 0.88               |             |                  |                       |                  |      |
| Flexible Discriminant Analysis (FDA)                                                         | Weighted response weights          | 0.95        | 0.9                                             | 0.93        | 0.85        | 0.55               | 0.86        | 0.89             | 0.99                  | 0.87             |      |
|                                                                                              |                                    |             | 0.99                                            | 0.87        |             | 0.60               | 0.86        |                  |                       |                  |      |
|                                                                                              |                                    |             |                                                 |             |             | 0.90               | 0.86        |                  |                       |                  |      |
| Resampling                                                                                   | –                                  | –           | 0.99                                            | 0.90        | –           | –                  |             |                  |                       |                  |      |
|                                                                                              |                                    |             | 20                                              | 0.87        |             |                    |             |                  |                       |                  |      |
|                                                                                              |                                    |             | 100                                             | 0.86        |             |                    |             |                  |                       |                  |      |
| Method                                                                                       | Polyreg                            | 0.93        | Polyreg                                         | 0.84        | Polyreg     | 0.87               |             |                  |                       |                  |      |
|                                                                                              | Bruto                              | 0.96        | Bruto                                           | 0.89        | Bruto       | 0.91               |             |                  |                       |                  |      |
|                                                                                              |                                    |             | Gen.ridge                                       | 0.89        |             |                    |             |                  |                       |                  |      |
| Surface Range Envelope (SRE)                                                                 | WRW                                | 0.92        | –                                               | –           | 0.99        | –                  | –           | 0.92             | –                     | –                |      |
|                                                                                              | Resampling                         |             | –                                               | –           |             | –                  | –           |                  | –                     |                  |      |
|                                                                                              | Quantile                           |             | 0.01                                            | 0.95        |             | 0.01,              | 0.98        |                  | 0.01                  | 0.95             |      |
|                                                                                              |                                    |             | 0.05                                            | 0.84        |             | 0.05               | 1.00        |                  | 0.05                  | 0.68             |      |
|                                                                                              |                                    |             | 0.1                                             | 0.69        |             | 0.1                | 1.00        |                  | 0.1                   | 0.69             |      |

(continued on next page)

Table 2 (continued)

| Algorithm                          | Configurable options      | Control AUC | Sensitive settings | Altered AUC | Control PPV | Sensitive settings | Altered PPV | Control Accuracy | Sensitive settings | Altered Accuracy |
|------------------------------------|---------------------------|-------------|--------------------|-------------|-------------|--------------------|-------------|------------------|--------------------|------------------|
| Classification Tree Analysis (CTA) | Weighted response weights | 0.91        | 0.90               | 0.88        | 0.86        | 0.25               | 0.85        | 0.89             | 0.66               | 0.88             |
|                                    |                           |             |                    |             |             | 0.66               | 0.82        |                  | 0.75               | 0.88             |
|                                    |                           |             |                    |             |             | 0.75               | 0.82        |                  | 0.90               | 0.87             |
|                                    |                           |             |                    |             |             | 0.90               | 0.82        |                  |                    |                  |
|                                    | Complexity parameter      |             | 0.025, 0.05        | 0.85        |             | 0.025              | 0.80        |                  | 0.025              | 0.85             |
|                                    |                           |             |                    |             |             | 0.05               | 0.83        |                  | 0.05               | 0.85             |
|                                    | Cross validation folds    |             | –                  | –           |             | –                  | –           |                  | –                  | –                |
|                                    | Minimum bucket            |             | 1000               | 0.9         |             | 1000               | 0.85        |                  | 1000               | 0.88             |
|                                    | Minimum split             |             | –                  | –           |             | –                  | –           |                  | –                  |                  |
|                                    | Maximum depth             |             | –                  | –           |             | –                  | –           |                  | –                  |                  |

occurrences. Any location with environmental conditions that falls within this envelope is included in the potential range for a species. To avoid the overpredictive effect of outliers, the envelope can be reduced to have limits at specified percentiles or standard deviations from the mean. Both FDA and SRE were tested for sensitivity to 3 configurable options.

*Classification Tree Analysis (CTA)* is a tree-based model. These models partition species data into increasingly homogeneous groups of presence or absence based on a sequence of decision rules (branches) to define their relationship to a set of environmental variables, the predictor variables. Each path through the tree, following different branches, arrives at a different environmental profile and probability of species presence. Each branch is defined by different ranges of one environmental predictor. The single classification tree is the most basic form of a decision tree model. As the name suggests, classification trees resemble a tree and consist of three different types of nodes, connected by directed edges (branches): root nodes, internal nodes and leaf nodes. CTA was tested for sensitivity to 6 configurable options within the BCCVL.

For each of the algorithms tested, different configuration options needed a different strategy for choosing values, depending on their type. For instance, configuration option settings involving positive real and natural numbers were both better assessed at increasing orders of magnitude, rather than on some linear scale. Some settings were categorical, e.g. whether to prune MARS is dichotomous. Some settings manage computational overheads, and are therefore only considered if the model provides poor fit.

Similarly to Hallgren et al. (2017), we used Australia-wide SDMs so as to be able compare results, in the case of the Koala sensitivity tests, with the published literature. Also in accordance with Hallgren et al. (2017) the experimental design for our sensitivity analyses is relatively simple; control models of Koala and Thorny Devil distribution were built using the default configuration option settings for each of the eight algorithms used in this study. Then, the algorithms were tested for their sensitivity to different settings of each of their configuration options that are configurable within the BCCVL. The method of choosing combinations of the configuration option settings for the sensitivity testing was the “one-at-a-time” or “OAT” method (Hamby, 1994). This involved changing one configuration option setting at a time for a particular algorithm, keeping all other configuration options for that algorithm at the default values. This method was chosen as it allows for the pinpointing of sensitivity to individual configuration options for each algorithm, so that our results can be used as a guide as to which configuration options require the most care when setting.

Only species occurrence records (presence points) for both the Koala and Thorny Devil were used in our sensitivity tests, as no true absence data was available for our study. Thus pseudo-absences were calculated in the BCCVL using the ‘random’ setting for this configuration option, which is the default in the BCCVL. All SDM experiments were

implemented with ecologically relevant domain constraints to encompass the observed distribution of Koala and Thorny Devil (a convex hull around the eastern, southeastern and southern distribution of Koalas, and a convex hull surrounding the observed distribution of Thorny Devil in the arid centre and west of the continent). This is to constrain the placement of pseudo-absence points generated by the BCCVL.

A process of expert elicitation was used to derive a range of appropriate values with which to test the sensitivity of our algorithms (Alkhairey et al., 2017); the same expert as for the Hallgren et al. (2017) study. For each algorithm, the first step in preparing elicitation (Low Choy et al., 2009, Ecology) was to identify the configurable option settings available through BCCVL. It was necessary to cross-reference the arguments required of the embedded call from the biomod package and the underlying R algorithms in multiple packages, as noted in the BCCVL provenance ([www.bccvl.org.au](http://www.bccvl.org.au)). This process was straightforward for several algorithms where the models, fitting and R settings are all clearly documented, in the same place: GLM, GAM, MARS, CTA, ANN and FDA (Hastie et al., 2009). Documentation for MaxEnt was dispersed in multiple publications (as detailed in Alkhairey et al., 2019 - unpublished manuscript).

Elicitation then proceeded following a similar process used for eliciting variable selection (Pirathiban et al., 2015), by prioritizing the settings, for inclusion in the sensitivity analysis, based on the expert's experience with using these algorithms (in contexts involving SDM or otherwise), and with SDMs in particular. Finally in the third stage, the expert was asked for numerical estimates, by nominating a “sensible” range of values for each settings, again based on their experience. These values were chosen to detect major shifts in the mapped predictions and performance of SDMs. The sensitivity analysis carried out in our study reflects a common “range-finding” situation with use of the virtual laboratory, where the modeller is initially exploring the wide range of possible settings values. Table 1 lists each algorithm and the configuration options that were tested in the BCCVL, as well as the value of the test settings. A brief description of the configuration options tested in this study for each of these algorithms is given in Appendix 1.

To evaluate the difference between the control SDMs (i.e. those using default configuration options settings) and altered-settings SDMs, we have visually assessed the results by comparing the geospatial distribution of the control SDM and the altered-settings SDM.

We have also noted the differences between the control model and altered-settings models in terms of how changing the configuration settings affects the Area under the Receiver Operating Characteristic (ROC) curve (AUC), and the Accuracy and Positive Predictive Value (PPV) statistics, all of which are calculated within the BCCVL when running a species distribution model (Table 2). The ROC plot is a graph with the False Positive Rate (1-Specificity) on the x-axis and the True Positive Rate (Sensitivity) on the y-axis plotted across the range of threshold probability values, where the area under the curve (AUC)

statistic is calculated by summing the area under the ROC curve. Area under the receiver-operator curve (AUC) is a threshold independent measure of predictive accuracy based on the ranking of locations. It is one of the most popular metric of model evaluation, and especially in the MaxEnt literature, and can be interpreted as the probability that “a randomly chosen presence location is ranked higher than a randomly chosen background point” (Merow et al., 2013; Jarnevich, 2015). The AUC can range from 0 to 1, although the practical range is from 0.5 to 1.0 since an AUC value of 0.5 represents a model that is no better than random (Jarnevich, 2015). The AUC represents a value that is used to evaluate the resulting SDM and to compare the performance among different algorithms, although essentially, we are using the AUC (and the Accuracy and PPV statistic) as a diagnostic tool to assess an algorithm's sensitivity to alternate configuration settings. ROC curves and the AUC statistic have been widely used in many fields such as medicine and psychology for many decades, and more recently in other areas such as machine learning, data mining and ecological modelling (Phillips et al., 2006; Santana et al., 2008).

A few authors have advocated calculating multiple evaluation statistics to assess model performance rather than relying on a single statistic, and we argue that the same approach should be used when evaluating the sensitivity of an algorithm, since each metric quantifies a difference aspect of predictive performance (or sensitivity) (Lobo et al., 2008; Jarnevich et al. 2015; Elith and Graham, 2009). The PPV is the proportion of correctly predicted presences, and the Accuracy statistic is the proportion of correctly predicted cases. These metrics were chosen among a range of statistical outputs generated by the BCCVL, mainly for their simplicity and ease of interpretation. Details of these evaluation metrics are in Appendix 2.

### 3. Results and Discussion

#### 3.1. Sensitivity shown by MaxEnt to configuration option settings

When modelling Koala distribution, the MaxEnt algorithm showed minor but noticeable sensitivity to the *weighted response weights* (WRW) setting throughout the entire east coast, and more so when settings were lower in value, compared to the BCCVL default. At higher WRW settings, the AUC value was marginally higher (0.95 compared to 0.94). When WRW was set to 0.25 (Fig. 1(a)), the (modelled) probability of Koala occurrence increased throughout the western extent of its range in New South Wales (NSW) and marginally in parts of Victoria (VIC), but decreased visibly throughout its range when WRW was set to 0.33. More marginal decreases in probability of occurrence resulted from values of 0.66 and 0.75 (see Table 1 for all test settings). MaxEnt also showed sensitivity to the *maximum number of iterations* option when set to lower-than-default values of 10 and 100 (Fig. 1(a)), showing slight decreases in the probability of Koala occurrence.

In the BCCVL, every feature threshold option (or ‘feature type’) is set to ‘yes’ by default. Sensitivity was tested by selecting ‘no’ for *linear* first, then adding a ‘no’ selection for *quadratic* (keeping *linear* as ‘no’). Then progressively adding the ‘no’ selection for *product*, *threshold* and *hinge* options, keeping all previous options checked as ‘no’. Decreased probability of occurrence resulted when the *linear* option was set to ‘no’, and when both the *linear* and *quadratic* options were both set to ‘no’ (Fig. 1(a)). Minor changes in distribution resulted from the combined *linear*, *quadratic* and *threshold* options being set to ‘no’, and slightly higher probability of occurrence, but negligible changes in extent, resulted from the combination of the *product feature threshold* being set to 40, the *quadratic feature threshold* being set to 5, and the *hinge feature threshold* being set to 8 (Table 1). The changes in Koala distribution seen as a result of these sensitivity tests are noticeable at regional levels in terms of probability of occurrence, although the overall spatial pattern of Koala distribution along the southern and eastern VIC and

eastern NSW does not change noticeably. This result corroborates Syfert et al. (2013) who found that the choice of feature type had negligible effects on predictive performance of their target species, at continental scale.

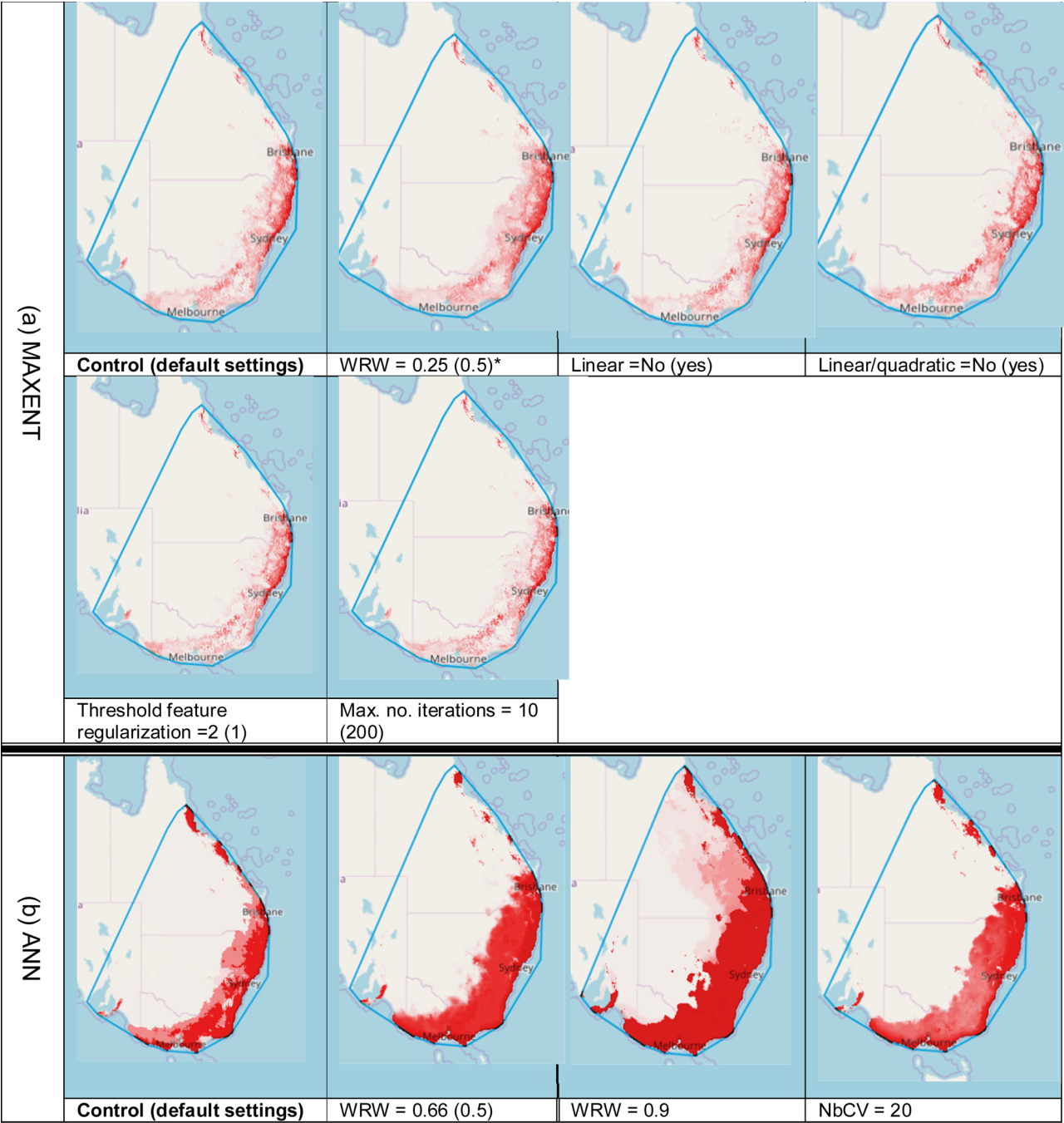
In terms of the Regularization options for MaxEnt, the *threshold*, *categorical*, *linear/quadratic/product* and *hinge feature regularization* options showed very minor sensitivity to altered settings (see for example the result for *threshold feature regularization* in Fig. 1). Anderson and Gonzalez (2011) found that their SDM results “varied greatly” according to the level of regularization, which indicates that stronger sensitivity was shown to this configuration option compared to that indicated by our study. The *threshold feature regularization* option shows changes in distribution that are discernable at regional levels in terms of changes to the probability of occurrence, with only very minor changes in range. Alternative values for the configuration settings only produced very minor changes to the AUC value, and the PPV and Accuracy statistics. The Accuracy statistic shows more sensitivity than the other statistics to changes in the configuration settings, with the most sensitivity being shown to the *threshold feature regularization*, the *linear/quadratic/product feature regularization*, and the combination of the *product*, *quadratic* and *hinge feature threshold* being set to 40, 5 and 8 respectively (Table 2).

It is clear that there were differences in the sensitivity to configuration settings that are attributable to the different species being modelled. The Thorny Devil sensitivity testing of MaxEnt revealed only very minor changes in distribution with lower values of the WRW setting (Fig. A1 in Appendix 3). Having the *prevalence*, *linear* and *quadratic threshold* and *linear*, *quadratic* and *threshold* settings set to ‘no’ also produced minor but still noticeable changes in distribution with the latter affecting the ‘smoothness’ of the distribution much more markedly in the Thorny Devil tests, compared to the Koala tests. Moreover, the Thorny Devil tests indicate some sensitivity when the threshold feature regularization is altered away from the control setting (unlike the case for the Koala tests), but minimal to no discernible sensitivity to changes in the value of the *categorical* and *hinge feature regularization*, nor when the *linear/quadratic* and *product feature regularization* options were changed together away from the default value (as was the case for the Koala tests). Of all the evaluation metrics, the AUC showed the least sensitivity to alternative configuration settings, showing minimal sensitivity to the *threshold feature regularization*, *linear&quadratic*, *linear&quadratic&threshold* and *prevalence* options. In contrast, the PPV and Accuracy statistics showed some, and often much greater, sensitivity to all configuration options (Table A1 in Appendix 3).

These results contrast with Convertino et al. (2014) whose sensitivity analysis results indicate that the *beta hinge (feature)* parameter tended to score highly in terms of factor importance, and *beta categorical*, the *maximum number of iterations*, *beta threshold*, *l2lq threshold*, *convergence threshold*, *hinge threshold*, and *beta multiplier* settings scored lower (in order of importance), according to the average habitat suitability result in Table 2 of Convertino et al. (2014).

#### 3.2. Sensitivity shown by ANN to configuration option settings

The ANN algorithm was tested for sensitivity to non-default values of WRW and the number of cross validations to find best size and decay parameters (NbCV). High sensitivity was seen to non-default values for both configuration options (Fig. 1(b)). In the case of WRW, only values higher than the control (> 0.5) resulted in altered distributions. There was both a moderately expanded range inland from southeast QLD to southeast SA (but a smaller range in northern QLD) and a higher probability of occurrence within that range for WRW values of 0.66 and 0.75, however with values of 0.9 and above, there was an expanded range all over, especially in eastern QLD. For NbCV, a value of 10 resulted in changes in the probability of occurrence in the southern parts of the



**Fig. 1.** Results of sensitivity testing of algorithms to alternative, i.e. non-default, configuration option settings, for the eight algorithms investigated in this study, for the Koala: (a) Maxent, (b) ANN, (c) GLM, (d) GAM, (e) MARS, (f) FDA, (g) SRE and (h) CTA. Selected results are shown here; only the results of sensitivity tests where algorithms showed more than minimal sensitivity to non-default configuration settings are shown. Values in brackets denote control (i.e. default) settings. \*The control value for each configuration option is stated once, in brackets beside the first test value for each configuration option shown, but not beside subsequent test values.

range, but a contracted range in northern QLD. For a value of 20, there was a slight expansion of range except in southeast QLD, which encompassed areas of slightly lower probability of occurrence. ANN showed minor sensitivity to altered values for the *WRW* and the *NbCV* settings in terms of the modelled distribution, and this was reflected in the evaluation statistics for the sensitivity tests, which showed mostly minor changes to the AUC, PPV and Accuracy values (Table 2).

The Thorny Devil sensitivity testing using ANN, produced results

that indicated a noticeable increase in distribution as *WRW* increased (changes only up to 0.66 were able to be determined for the Thorny Devil tests). The modelled distribution of Thorny Devil noticeably decreased as *NbCV* increased from 10 to 20, whereas in the Koala sensitivity tests the modelled distribution noticeably increased as *NbCV* increased (Fig. A1, Appendix 3). In addition, the Thorny Devil sensitivity tests of ANN showed a noticeably larger impact on the evaluation metrics, with the AUC, PPV and Accuracy all showing higher sensitivity



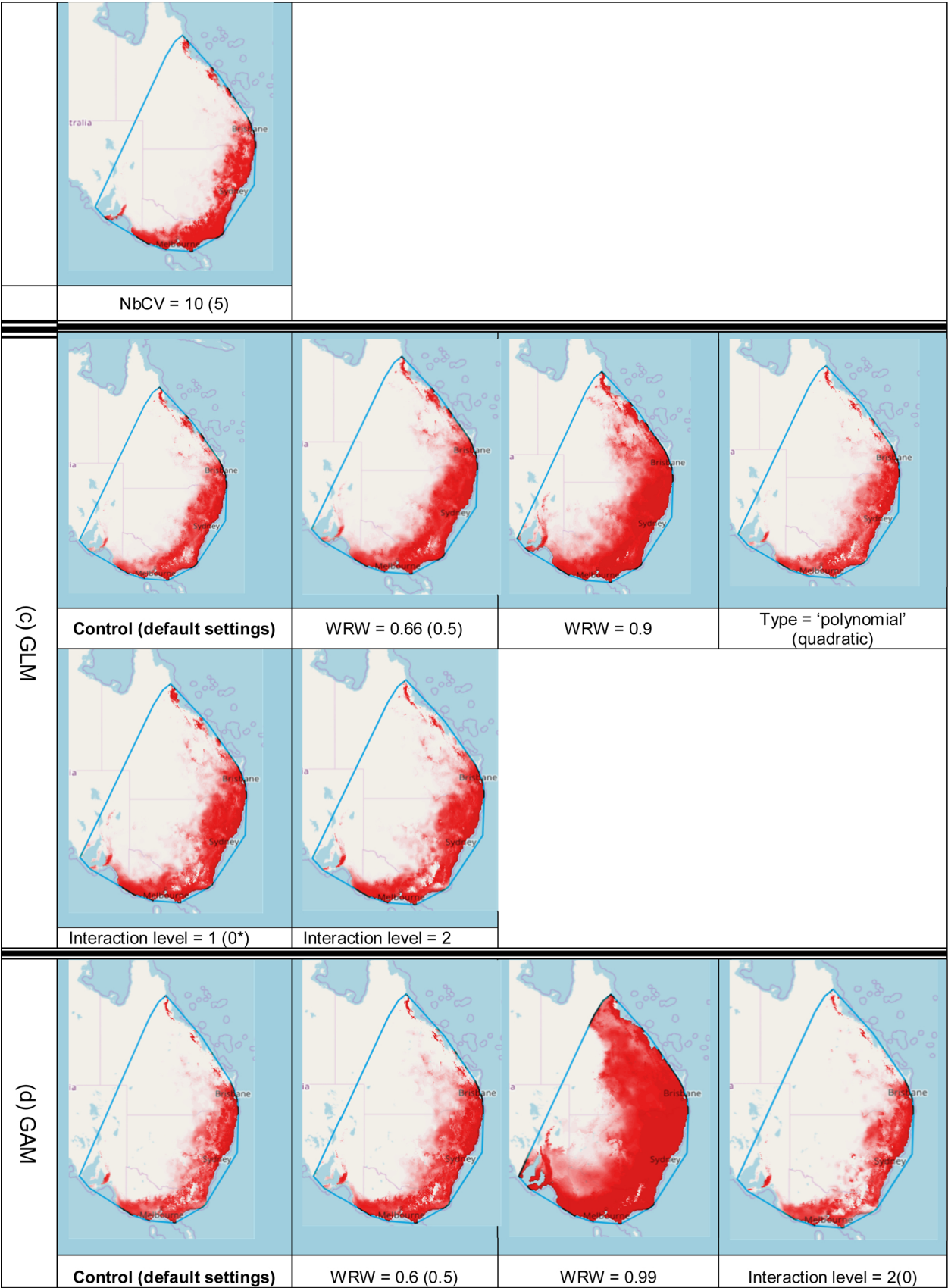


Fig. 1. (continued)

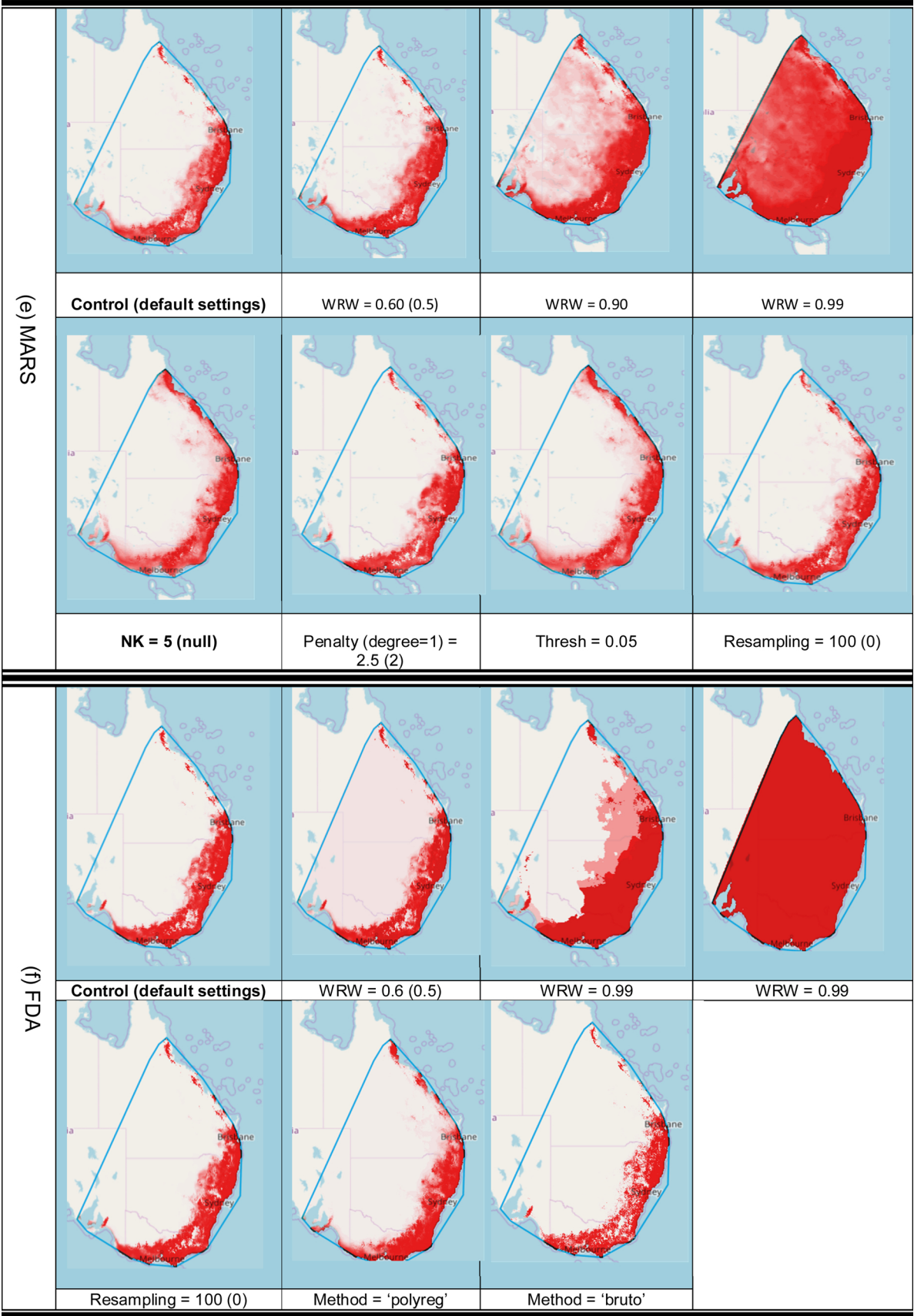


Fig. 1. (continued)

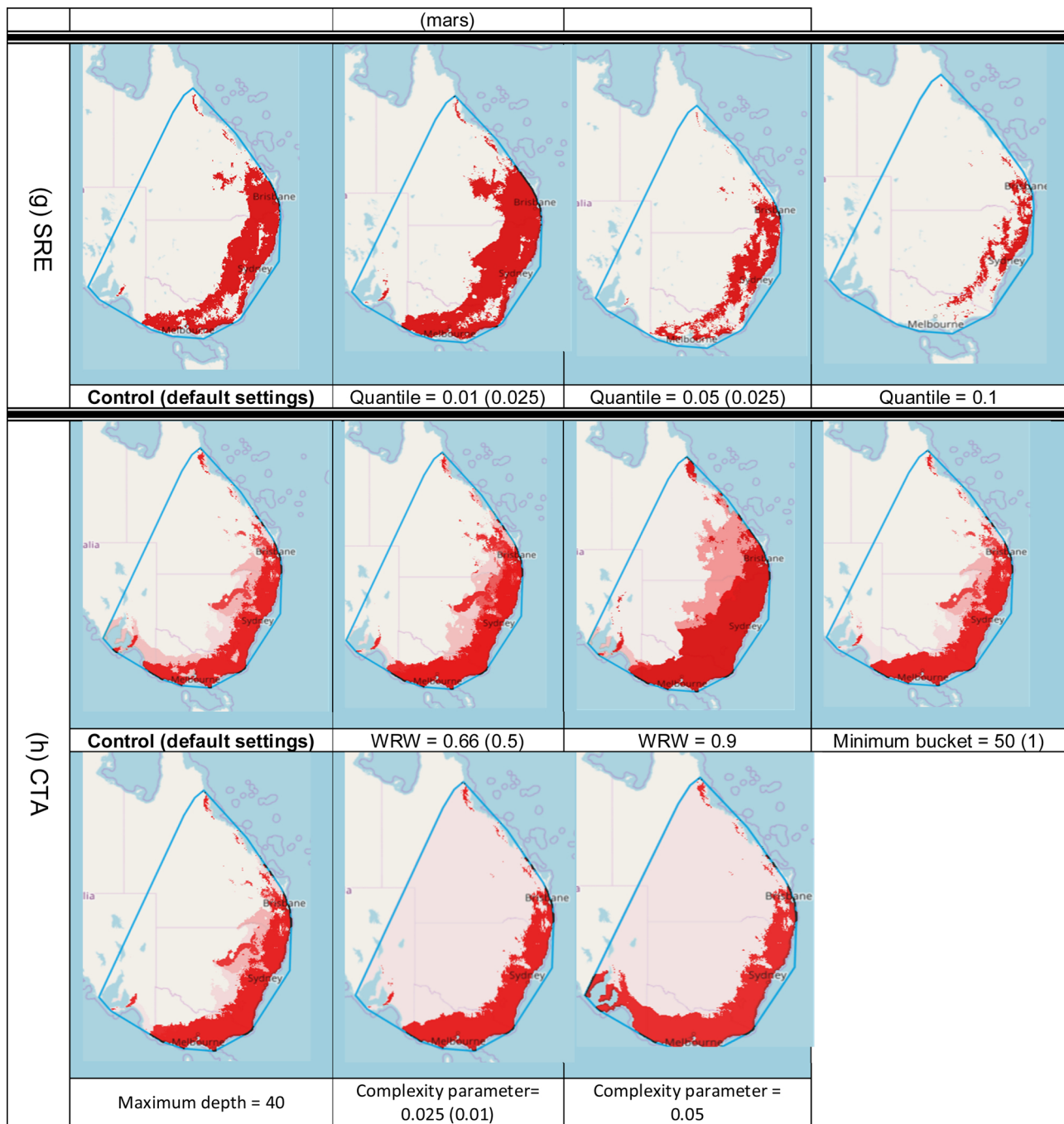


Fig. 1. (continued)

to changes in the configuration settings compared to the Koala sensitivity tests (Table A1, Appendix 3). This indicates that the sensitivity of ANN to configuration options differs according to the species modelled.

### 3.3. Sensitivity shown by GLM to configuration option settings

The GLM algorithm was tested for sensitivity to non-default values of the *WRW*, *type*, *interaction level* and *test* configuration options. High sensitivity was seen in the modelled distribution maps to non-default values for the *WRW* option, with greater sensitivity the higher the value for this setting (Fig. 1(c)). Considerable sensitivity was also shown to other settings for the *type* configuration option, with the *polynomial*

setting resulting in a more westward distribution of the Koala population (Fig. 1(c)). GLM showed strong sensitivity to the interaction level, with greater values resulting in expanded Koala populations in some areas (northern NSW, VIC) and contracted population/lower occurrence probability in other areas (southern NSW, along the northern QLD coast, parts of eastern VIC) (Fig. 1(c)). No noticeable sensitivity was shown to alternative settings of the *test* configuration option. Altered values for the *type* and the *interaction level* settings produced minor increases in the AUC and Accuracy values, and slightly larger increases in the PPV values (Table 2), indicating generally low sensitivity of GLM according to these evaluation metrics, for the Koala sensitivity tests.

The Thorny Devil tests indicate that GLM shows more sensitivity to changes in configuration settings than is the case in the Koala tests; this is true for the *WRW*, *type* (particularly for the *simple type* setting), *interaction level*, and *test* settings, with the *BIC* setting for the *test* configuration option showing noticeably different distribution compared to the control (*AIC*) setting (Fig. A1). The evaluation statistics for the Thorny Devil sensitivity tests indicates that GAM is sensitive to all four configuration options tested according to the AUC and Accuracy statistic, both of which, for example, decrease with increasing *WRW* values. The PPV metric indicates sensitivity only to the *type* and *interaction level* configuration options (Table A1). The metrics we investigated corroborate the modelled distribution results in showing that GLM shows more sensitivity when tested with the Thorny Devil dataset than with the Koala dataset.

### 3.4. Sensitivity shown by GAM to configuration option settings

The GAM algorithm was tested for sensitivity to non-default values of numerous configuration options. The change in Koala distribution increased in terms of probability of occurrence when the value of the *WRW* was increased to 0.6 from the control value of 0.5. Values of 0.9 and 0.99 resulted in not only increases in the probability of occurrence but also a larger extent of Koala distribution inland from the observed distribution in QLD, NSW, VIC and South Australia (SA) (Fig. 1(d)). As values of the *Interaction level* setting increased from the control to 1 and 2, there were areas within the Koala distribution which showed lower probability of occurrence, and at a value of 2, there is a contraction of Koala range in southeast NSW and eastern VIC (Fig. 1(d)). Changes in the value of the *Epsilon* setting produced only very minute, almost imperceptible changes in Koala distribution. There were no discernible changes in Koala distribution for test values of *irls.reg* from 0.1 to 0.9; *Max MLE iterations* from 50 to 200; *Convergence tolerance* from 1 E-5 to 1 E-6; and *Number of halvings* from 5 to 30. Only alternative higher-than-default values for the *WRW* configuration option resulted in a very small increase in the PPV and Accuracy statistic, and the AUC indicated no sensitivity to any alternative configuration values, for the Koala sensitivity tests (Table 2).

For the Thorny Devil sensitivity tests, minimal (but perhaps locally significant) sensitivity was shown to increases in *WRW* up to 0.6 (sensitivity to higher values was indeterminable). Some sensitivity was shown to *interaction level* (from 1 to 2), particularly in northwest WA, but noticeably more sensitivity was shown by GAM to increasing values of *irls.reg* and *Epsilon* in the Thorny Devil sensitivity tests compared to that shown in the Koala tests: the Thorny Devil sensitivity tests resulted in quite minimal but still noticeable sensitivity of GAM to increases in the value of *max MLE iterations*, *convergence tolerance* and *number of halvings* (Fig. A1). The evaluation statistics for the Thorny Devil tests indicate GAM is sensitive to altered values for all of the configuration settings tested according to the AUC and PPV statistic (the latter showing more sensitivity). The Accuracy statistic showed less sensitivity, both in terms of the number of configuration options, and the magnitude of sensitivity shown (Table A1, Appendix 3). Certainly these statistics reveal that GAM is more sensitive to alternative configuration settings when modelling the distribution of the Thorny Devil compared to the distribution of the Koala.

### 3.5. Sensitivity shown by MARS to configuration option settings

MARS sensitivity testing showed a higher probability of occurrence and expanded range with higher values of *WRW* than the default (at 0.6, there is higher probability of occurrence within the range produced by the default, and patches of higher probability of occurrence outside that range; at 0.9 and above, there is a much expanded range with moderate probability of occurrence) (Fig. 1(e)). Moderate sensitivity was shown by MARS to non-default values of *NK* (maximum number of terms). There was a decreased probability of occurrence at the western

edge of the range and an expanded range in northern QLD when *NK* was set to 5 and 10 (Fig. 1(e)), and minimal changes resulting from settings of 15, 20 and 100. Minimal but still noticeable sensitivity was seen to the *thresh* option when it was set to 0.01 and 0.005, however, the eastern and southern population contracted, and northern population expanded with higher values of 0.05.

Values for *penalty* which were lower than the default elicited some sensitivity; when set to 2.5, there is a slightly expanded range in northeast NSW and slightly less in the high country in NSW/VIC (Fig. 1(e)). For extremely high values for resampling (i.e. when set to 100), the Koala distribution shows a slightly altered range in NSW and a slightly expanded range in QLD (Fig. 1(e)). No noticeable sensitivity was shown to the *prune* setting. Altered values for the *NK* and the *thresh* settings produced decreased AUC, PPV and Accuracy values (Table 2). Increased values of the *WRW* option caused decreased values for all these metrics. Overall the PPV statistic was the most sensitive to changes in the configuration options tested, and AUC the least sensitive metric, in terms of the number of options that showed sensitivity and the magnitude of the sensitivity shown.

The Thorny Devil tests indicate that MARS shows more sensitivity to changes in some configuration settings than the Koala tests; there is considerable sensitivity shown to altered values of the *NK* option, with low values (*NK* = 5) increasing the area which has any probability of distribution of Thorny Devil markedly, and with high values (*NK* = 100) reducing the extent of modelled Thorny Devil occurrence. There is also noticeable sensitivity to *penalty* setting with an increasing extent of occurrence as values increase (which is not the case for the Koala). High sensitivity was shown by MARS to the *thresh* option, with increasing extent of occurrence with higher values, similarly to the Koala tests, but the sensitivity was higher in the Thorny Devil tests. As with the Koala tests, MARS showed high sensitivity and a larger extent of occurrence to increasing values of *WRW*, and more sensitivity to increasing values of *resampling* than the Koala tests indicate (Fig. A1, Appendix 3). The evaluation statistics for the Thorny Devil sensitivity tests indicate that MARS shows more sensitivity to configuration settings when simulating the Thorny Devil compared to the results of the Koala sensitivity tests, with more configuration options showing a greater magnitude of sensitivity to altered settings, and in the case of MARS, it is the PPV and AUC statistics which indicate greater sensitivity, compared to the Accuracy statistic (Table 1 Appendix 3). The *NK* and *Thresh* options showed the greatest sensitivity across all three metrics in the Thorny Devil tests.

### 3.6. Sensitivity shown by FDA to configuration option settings

For the Koala sensitivity testing of the FDA algorithm, very high sensitivity was seen to higher-than-default values for *WRW*, with both higher probability of occurrence and a much larger extent of distribution above *WRW* values of 0.6 or higher (Fig. 1(f)). FDA showed moderate sensitivity to alternative values of the *method* configuration option, in terms of the probability of occurrence (compared to the predicted range) resulting from the default setting. The *polyridge* setting for the *method* option resulted in higher probability of occurrence and for the *bruto* setting they resulted in lower probabilities of occurrence (Fig. 1(f)). Negligible sensitivity was seen for alternative values of the *resampling* option at the continental scale. Altered values for the *WRW* and the *method* settings produced both increased and decreased AUC, PPV and Accuracy values (Table 2). The PPV statistic showed higher sensitivity to these, and was also quite sensitive to the *resampling* configuration option. The Thorny Devil sensitivity tests revealed high sensitivity in the modelled distributions to changes in the value of *WRW*, *resampling* and *method* configuration options; with the latter two showing more sensitivity in the Thorny Devil tests compared to the Koala tests (Fig. A1). The evaluation statistics also revealed high sensitivity to changes in the value of the three configuration options, and showed higher sensitivity than the evaluation statistics from the Koala



sensitivity tests, particularly the *resampling* and *method* options (Table A1).

### 3.7. Sensitivity shown by SRE to configuration option settings

The SRE algorithm was tested for sensitivity to alternative values of the *WRW*, *resampling* and *quantile* configuration options. High sensitivity was seen to non-default values for the *quantile* option, with an expanded Koala range seen at a value of 0.01, and an increasingly contracted range seen for values of 0.05 and 0.1 (Fig. 1(g)). No noticeable sensitivity was shown to the *WRW* and *resampling* settings. These results were reflected in the evaluation statistics, with only the *quantile* option exhibiting any sensitivity; increasing *quantile* values resulted in decreasing values of AUC and Accuracy but higher values of PPV (Table 2). There was a similar decrease in the extent of modelled Thorny Devil distribution with increasing values of the *quantile* configuration setting, as for the Koala (Fig. A1). No change in distribution was seen with increasing values of 'resampling' or *WRW*. The *quantile* setting showed a similar pattern of sensitivity in the evaluation statistics to that seen in the Koala sensitivity tests, however the Thorny Devil tests also indicated sensitivity to the *WRW* setting, in terms of altered AUC, PPV and Accuracy values, and also sensitivity to the *resampling* setting in terms of altered AUC values. These results indicate the SRE algorithm shows more sensitivity when tested with the Thorny Devil dataset, compared to the Koala dataset (Table A1).

### 3.8. Sensitivity shown by CTA to configuration option settings

The CTA algorithm's modelled distribution showed high sensitivity to the *WRW* option (Fig. 1(h)), with higher values resulting in an expanded Koala range and higher probability of occurrence in many areas within that range. High sensitivity was also shown to changes in the setting of the *complexity parameter* option (showing a much increased probability of Koala occurrence in western VIC when the value is 0.025, and an increase in range and probability of occurrence again when the value is 0.05 (Fig. 1(h)), compared with the default value of 0.01). CTA was also moderately sensitive to different settings of the *maximum depth* option, particularly when the value is set to 40 (i.e. higher than the default) (Fig. 1(h)). The CTA algorithm showed minor sensitivity to the *minimum bucket* setting; there was a minor decrease in the probability of occurrence in western Victoria when *minimum bucket* was set to 50, but no other alternative settings produced different distributions from the control (20). No noticeable sensitivity was seen to the *cross-validation folds* and *minimum split* configuration options. All evaluation statistics showed sensitivity to the *WRW*, *complexity parameter* and *minimum bucket* settings (Table 2). The PPV statistic showed more sensitivity to alternative configuration settings than either the AUC or Accuracy statistic, particularly the *WRW* setting.

The Thorny Devil tests indicate that CTA shows more sensitivity to changes in configuration settings than is the case in the Koala tests; this is true for *WRW*, *cross validation folds*, *minimum bucket*, *minimum split*, and *maximum depth* configuration options. The Thorny Devil tests indicate high sensitivity of CTA to the *complexity parameter* (as with the Koala tests) (Fig. A1). The CTA algorithm also showed more sensitivity in the Thorny Devil sensitivity tests than the Koala tests, in terms of the impact of the alternative configuration settings on the evaluation statistics. In these tests the AUC exhibited greater sensitivity than either the PPV statistic (in terms of the number of options that showed sensitivity) or the Accuracy statistic (in terms of the magnitude of the sensitivity displayed by most of the configuration options (Table A1).

## 4. Summary and conclusions

Our study aimed to comprehensively test the sensitivity of eight algorithms that are widely used in species distribution modelling to variation in their configuration settings. In addition to the two

algorithms which were tested for sensitivity to configuration settings in terms of differences in Koala distribution in Hallgren et al. (2017), there are now eight more SDM algorithms which have been comprehensively tested and have been shown to exhibit sensitivity, in terms of how differently configured algorithms can result in different (and often remarkably different) modelled distributions and evaluation metrics for two species, the Koala and the Thorny Devil.

Our results indicate that for many of the algorithm tested in this study, not all configuration options show the same sensitivity, as might be expected. Of the eight algorithms tested, the most flexible ones (MARS, CTA, FDA, GAM, SRE), showed greater sensitivity to more of their alternative configuration settings which were tested in this study, in terms of changes in the modelled probability of occurrence of the species (Koala or Thorny Devil) or the range in the species distribution. This is understandable given that many of these settings govern the flexibility, in terms of how closely the model fits the data (Hastie et al., 2009). MaxEnt and GLM have fewer configuration settings, but also showed less sensitivity to their settings. This can be explained since these approaches are more model-based (Austin, 2002), so that rather than having many small decisions regarding model flexibility, there are fewer major decisions on model complexity. These can be more strongly related to ecological theory (e.g. Austin, 2002; Low-Choy and Huijbers, 2017), meaning algorithm sensitivity can be under the control of the modeller with some ecological knowledge.

Our research shows that for many of the configuration options investigated, particularly those for MARS, CTA, SRE, ANN and FDA, the choice of settings for these options can impact the resulting modelled distribution markedly, as well as the AUC, PPV and Accuracy statistics (it is likely that other evaluation statistics would also show some sensitivity, but this needs empirically verifying). Hence, appropriate care must be taken to choose sensible and justifiable values for these algorithms.

Our research also indicates that algorithm sensitivity to configuration settings differs according to the species modelled, and that this differential sensitivity is greater in some algorithms than others. Algorithm sensitivity is also somewhat dependent on type of model output (distribution map/evaluation metric) used to diagnose this sensitivity. Some evaluation statistics reveal more or less sensitivity of the algorithm to alternative configuration settings. Our finding of the differential sensitivity of the different evaluation statistics to alternative SDM configuration settings, and this justifies our use of more than one them in this study to evaluate the sensitivity of the algorithms. This aspect of our results supports a growing chorus which indicates that it is not wise to rely on the AUC as a measure of model skill (e.g., Townsend Peterson et al., 2008; Lobo et al., 2008).

One of the aims of our study was to draw conclusions as to how the sensitivity of SDM algorithms may impact on the reliability of SDM results, given the frequently unjustified nature of the default algorithm settings, and the generally low level of awareness of, and attendance to, this issue in much of the published literature (keeping in mind that this reliability is determined by a number of factors including the algorithm chosen, the available occurrence data and the objective of the modelling study).

With respect to an algorithm's configuration, we suggest that SDM studies that have not carefully chosen configuration settings may yield results that are inaccurate or inappropriate to the desired ecological scale, to the extent that the algorithm is sensitive to those settings. At the very least, this source of uncertainty should be acknowledged in published SDM studies.

Another aim of this study was to define which configuration options need setting thoughtfully, given an algorithm's sensitivity to them, in terms of how the projected distribution was affected. We found that all algorithms tested were sensitive to sampling weights, particularly the *WRW* configuration option, except SRE. This sensitivity confirms that the samples provided are not exhaustive of the complex Australian environment, and sampling effort appears to have varied with



environment. This is important since it signals to BCCVL users that they need to check the data they are putting into SDMs in terms of the sampling weights (which means changing the WTW setting, for the algorithms are very sensitive to this).

Our results suggest that for MaxEnt, care must also be taken when configuring the *linear* and *quadratic* settings, the *threshold feature regularization* and the *maximum number of iterations*, due to the sensitivity of the modelled distribution to these configuration options. For ANN, care should be taken when setting the *NbCV* option; for GLM, the configuration options to set most carefully are the *type* and *test* settings, and for GAM, care must be taken when setting the interaction level. The most sensitive configuration options for the MARS algorithm that need to be set carefully are the *resampling*, *prune*, *NK*, *penalty*, and *thresh*; for FDA, the *resampling* and *method* configuration options require careful setting. For SRE, the *quantile* setting is the most sensitive and therefore needs most care when setting, and for CTA, care should be taken when setting the *minimum bucket* and the *maximum depth* configuration options.

We believe our results demonstrate unequivocally the importance of the configuration settings of algorithms used for species distribution modelling given the sensitivity to such settings. Furthermore, the documentation of these settings is necessary to increase the scientific robustness, transparency and reproducibility of species distribution modelling studies (as noted in Hastie et al., 2009). While only a brief description of the configuration options tested in this study is given in Appendix 1, in light of our results, we believe that it is necessary to

have a good understanding of an algorithms configuration options, in order to be able to justify the settings used, and to understand the impact each configuration setting has on the model and resulting species distribution and evaluation statistics. We note that all sensitivities are relative to the control model chosen and are also dependent upon the test values chosen, which is why great care was taken to choose test values based on expert knowledge of the algorithms, and which were numerically and statistically appropriate for the algorithm tested. Nevertheless, for the configuration option values to which certain algorithms did not show much sensitivity to, there remains the possibility that sensitivity could potentially be shown to a broader range of test values.

Lastly, while the sensitivity we have documented to the algorithms we have investigated in this study may not hold for other locations or species, we have demonstrated that our methodology can be used to effectively and efficiently investigate the sensitivity of SDM algorithms to their configuration settings for any SDM study.

## Acknowledgments

The BCCVL is supported by the Australian Research Data Commons (ARDC). The ARDC is supported by the Australian Government through the National Collaborative Research Infrastructure Strategy (NCRIS). Griffith University provided the research funding to conduct this study. No co-authors have any conflicts of interests to declare.

## Appendix 1 Configuration option descriptions for algorithms used in the sensitivity analysis testing. Descriptions obtained from literature (where cited), or the bccvl.org.au

| Algorithm                          | Configuration options         | Description of configuration options                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classification Tree Analysis (CTA) | Weight response weights (WRW) | Allows users to give more or less weight to particular observations. If the default setting is used (i.e. "NULL"), each observation (presence or absence) has the same weight, regardless of the number of presences and absences. If the value is set below 0.5 absences are given more weight, whereas a value above 0.5 gives more weight to presences.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                    | Resampling                    | The number of permutations to estimate the importance of each variable. If a value larger than 0 is used, the algorithm will produce an object called 'variableImportance.Full.csv', high values of which mean that the predictor variable has a high importance, whereas values close to 0 indicate that the predictor variable has no importance.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                    | Method                        | The method to be used to fit the decision tree (classification or regression). For categorical data such as the presence/absence of species, the 'class' method should be used, which is the default for this option.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                    | Cross-validation folds        | The function of cross-validation folds that should be created for training (cv - 1) and testing (1). The default number of cross-validations is 10, which means that the datasets is divided into 10 subsets, using 9 subsets as training data to calibrate the model, and 1 subset as test data. This process is repeated for all possible combinations of training and test subsets (a total of 10 times), and then the error rates can be used to prune the tree using the standard '1-SE' rule, which states that the tree with the most nodes should be selected as long as the corresponding cross-validation error is within one standard deviation of the minimum cross validation error. The corresponding complexity parameter value can then be specified to generate an optimally pruned tree. |
|                                    | Minimum bucket                | The minimum number of observation that needs to be present in any terminal node.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                    | Minimum split                 | The minimum number of observations that must exist in a node for a split to be attempted. This is a good way to limit the growing of the tree. When a node contains too few observations, further splitting will result in overfitting. The optimum minimum split depends on the number of observations and predictor variables in a dataset. With a limited number of observations and predictor variables, stopping early will result in no tree at all. With a lot of observations, a CTA can stop early and still obtain a large enough decision tree. The more predictor variables in a dataset, there is a larger possibility of having some accidental relationship between one of the variables and the response. So with a lot of variables it is advisable to stop earlier.                      |
|                                    | Complexity parameter          | The complexity measure is a combination of the size of a tree and the ability of the tree to separate the classes of the response variable. If the next best split in growing a tree does not reduce the tree's overall complexity by the value of this option, the algorithm will terminate the growing process. A complexity parameter value of 1 will always result in a tree with no splits, while a value of 0 will build a tree with the maximum possible complexity, which can potentially be very complex.                                                                                                                                                                                                                                                                                         |
|                                    | Maximum depth                 | The maximum number of splits of the tree. This is a limit to stop further splitting of nodes when the specified tree depth has been reached during the building of the initial decision tree.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                    | Random seed                   | Random seed is a number which is used to generate a set of pseudorandom numbers by some algorithms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                         |                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Generalized Linear Model (GLM)</b>   | WRW                                                                                                                                                                                                                                    | As for CTA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                         | Resampling                                                                                                                                                                                                                             | As for CTA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                         | Type                                                                                                                                                                                                                                   | The type of regression model to use. This can be either a linear (option 'simple'), a quadratic or a polynomial model. The default for this option is 'quadratic' which creates a curved model with one 'hump', i.e., a U or inverted U shape.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                         | Interaction level                                                                                                                                                                                                                      | The number of interactions between predictor variables that need to be considered. A value of 1 will fit an additive model, a value of 2 will result in a model with 2-way interactions, etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                         | Test                                                                                                                                                                                                                                   | The biomod2 package uses an automatic stepwise selection procedure, which means that the model is built by sequentially adding or dropping predictor variables and testing whether they improve the fit of the model. Predictors that do not improve the fit of the model will be dropped. The Test option indicates which criteria should be used to test the fit of the model; the Akaike Information Criterion (AIC), which is the default, or the Bayesian Information Criteria (BIC). By selecting 'none', the stepwise procedure will be switched off resulting in a model in which all predictor variables are retained.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | Family                                                                                                                                                                                                                                 | The description of the error distribution of the response variable and the link function used in the model. For binary data such as presence/absence of species, the binomial family should be used, which is the default for this option.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                         | Mustart                                                                                                                                                                                                                                | Starting values of the vector of means.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                         | Epsilon                                                                                                                                                                                                                                | The positive convergence tolerance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                         | Maximum MLE iterations                                                                                                                                                                                                                 | The maximum number of IWLS iterations to find the maximum likelihood estimates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                         | MLE iteration output                                                                                                                                                                                                                   | Determines whether output should be produced for each IWLS iteration. The default is 'no', but if 'yes' is selected, the Rout log file in the BCCVL SDM experiment results will show the fit of the model for each iteration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Artificial Neural Network (ANN)</b>  | Random seed                                                                                                                                                                                                                            | As for CTA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                         | WRW                                                                                                                                                                                                                                    | As for GLM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                         | NbCV                                                                                                                                                                                                                                   | The number of cross-validations. Cross validation processes the data by dividing it into training sets and test sets. Each training set is used once to train the data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Maximum Entropy model (MaxEnt)</b>   | WRW                                                                                                                                                                                                                                    | As for CTA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                         | Maximum number of iterations                                                                                                                                                                                                           | The number of iterations of the optimization algorithm after which training is stopped.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                         | Linear/quadratic/product/threshold/hinge feature thresholds                                                                                                                                                                            | This set of configurations include 5 different features of MaxEnt: <ul style="list-style-type: none"> <li>• The linear feature class ensures that the mean value of a predictor (e.g. rainfall) where the species is predicted to occur approximates the value the mean value for where it is observed to occur (Merow et al., 2013). Used when a species niche is truncated; one side of the unimodal curve is not part of the background sample.</li> <li>• A quadratic feature constrains the variance in a predictor where the species is predicted to occur to match observation (Merow et al., 2013).</li> <li>• A product feature constrains the covariance of a predictor with other predictors. It is equivalent to interaction terms in regression, when linear features are also included (Merow et al., 2013).</li> <li>• Threshold features make a continuous predictor binary by generating a feature whose value is '0' below the threshold and '1' above (Merow et al., 2013). Used when a physiological tolerance limit exists.</li> </ul> Hinge features are like threshold features except that a linear function is used instead of a step function (Merow et al., 2013). Used when complex but smoothed response curves (like GAM) are produced. |
|                                         | Regularisation options: <ul style="list-style-type: none"> <li>- threshold feature</li> <li>- categorical feature</li> <li>- linear/quadratic/product feature</li> <li>- hinge feature</li> <li>- Regularization multiplier</li> </ul> | This parameter is denoted by lambda in Equations 3 & 4 in Elith et al. (2011, Diversity and Distributions) and also explained (denoted in beta) in Equation 9 of Merow, Smith and Silander (2013). It is the penalty for over-parameterisation in the gain function (related to the likelihood ratio of the average presence to the average background point) for the size of the coefficients that accompany feature functions (Phillips et al., 2006; Phillips and Dudík, 2008) in the linear predictor. Higher values mean that there is more penalty on the size of coefficients, and hence penalizes feature functions that exhibit greater variability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                         | Prevalence                                                                                                                                                                                                                             | Overall prevalence of the species throughout the study region: i.e. probability of presence at ordinary occurrence points. It depends on resolution and on the study region. In what proportion of rasters is the species presumed present. Enables inverting probability output by MaxEnt,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Generalized Additive Model (GAM)</b> | WRW                                                                                                                                                                                                                                    | As for GLM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                         | Interaction level                                                                                                                                                                                                                      | As for GLM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                         | irls.reg                                                                                                                                                                                                                               | The size of the ridge regression penalty to the model to impose identifiability; this should be 0 for most models. The iteratively re-weighted least squares method by which GAMs are fitted can fail to converge in some circumstances.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                         | Epsilon                                                                                                                                                                                                                                | As for GLM; Used for judging the conversion of the GLM IRLS loop in GAM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                         | Max MLE iterations                                                                                                                                                                                                                     | Maximum number of IWLS iterations to find the maximum likelihood estimates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                         | Convergence tolerance                                                                                                                                                                                                                  | The convergence tolerance parameter to use in the GCV/UBRE optimization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                         | Number of halvings                                                                                                                                                                                                                     | If a step of the GCV/UBRE optimization method leads to a worse GCV/UBRE score, then the step length is halved; this is the number of halvings to try before giving up.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                        |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Surface Range Envelope (SRE)</b>                    | WRW                    | As for GLM                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                        | Resampling Quantile    | As for CTA<br>The quantile used to remove the most extreme values of each environmental variable for determining tolerance boundaries; the default is set to 0.025, which corresponds to a 95% confidence interval in normal distributions.                                                                                                                                                                                                          |
| <b>Flexible Discriminant Analysis (FDA)</b>            | WRW                    | As for GLM                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                        | Resampling Method      | As for CTA; Number of permutations to estimate the relative importance of each variable.<br>As for CTA; the regression method used in optimal scaling.                                                                                                                                                                                                                                                                                               |
| <b>Multivariate Adaptive Regression Splines (MARS)</b> | NK (max. no. of terms) | The maximum number of terms in the model before pruning occurs.<br>In MARS, there have a set M to collect the best production pair. This configuration is used to limit the size of set M.<br>$\hat{\beta}_{M+1}h_1(X) \cdot (X_j - t)_+ + \hat{\beta}_{M+2}h_1(X) \cdot (t - X_j)_+, h_1 \in M$ $h(X) = (X_j - t)$<br>“t” is the value of a knot in every piecewise;<br>MARS will add the products to M set until the number = NK (which we preset) |
|                                                        | Penalty                | The generalized cross validation penalty per knot. MARS uses generalized cross-validation (GCV) to prune some terms to improve the result accuracy. The Prune is used to limit GCV;<br>$GCV(\sigma) = \frac{\sum_{i=1}^N (y_i - \hat{f}_\sigma(x_i))^2}{(1 - M(\sigma)/N)^2}$ $M(\sigma) = r + cK$<br>C is Penalty. (Trevor et al., 2009)                                                                                                            |
|                                                        | Thresh                 | Forward stepwise stopping threshold; the forward pass terminates if adding a term changes Rsq by less than the threshold.                                                                                                                                                                                                                                                                                                                            |
|                                                        | Prune                  | In this experiment in BCCVL, there are two configurations:<br>1 Prune method: including backward, no prune, exhaustive, forward, seqrep and CV to Prune terms to improve the accuracy.<br>Prune model terms: preset the maximum size of set M after prune.                                                                                                                                                                                           |
|                                                        | WRW                    | As for GLM                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                        | Resampling             | As for CTA                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                        |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Appendix 2 Further details of the formulation of the evaluation metrics used to diagnose the sensitivity shown by algorithms to alternative configuration values

1) The Positive Predictive Value (PPV) is the proportion of correctly predicted presences:

$$PPV = TP / (TP + FP), \quad (1)$$

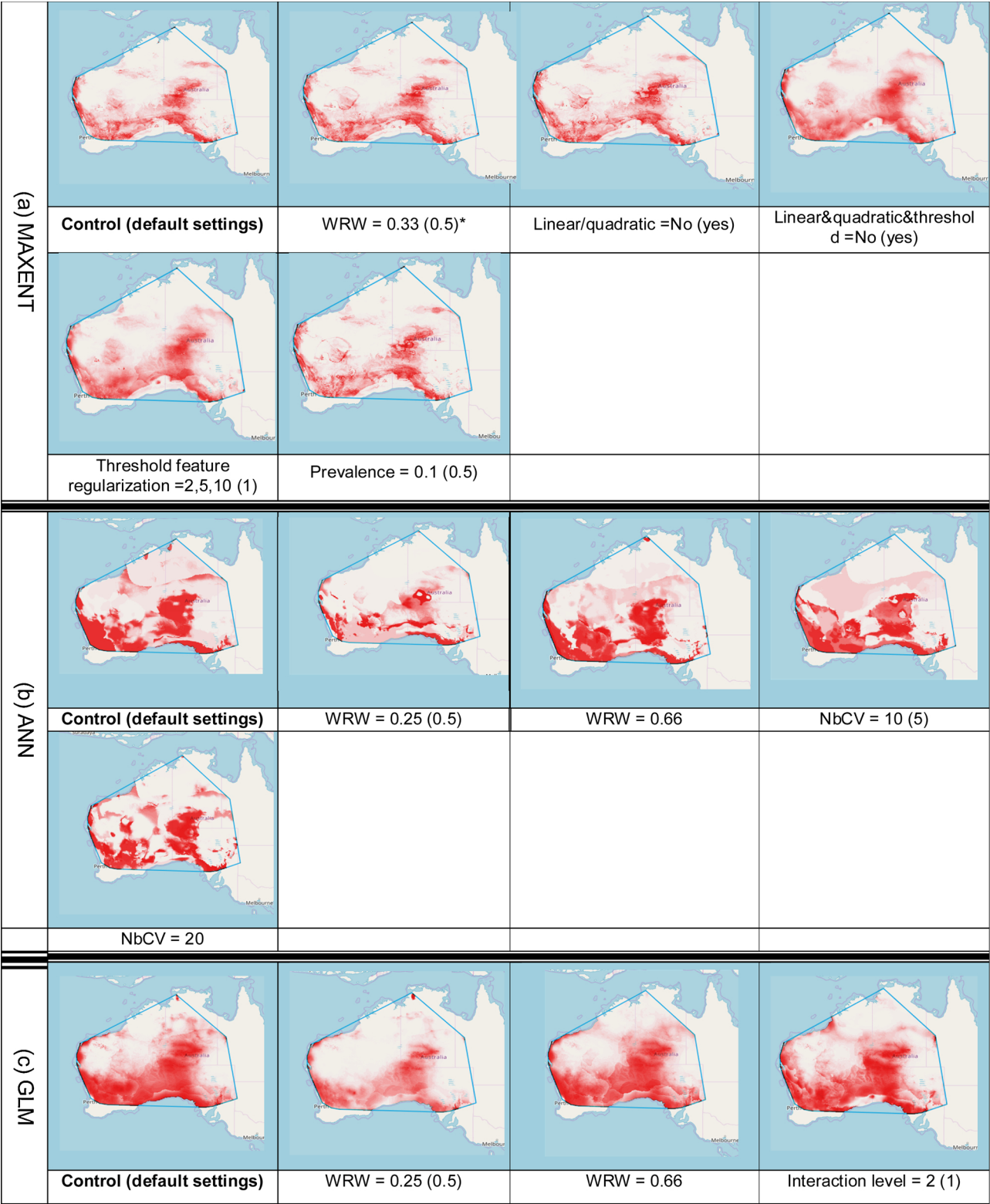
Where TP is ‘true presences’ or the number of presences which are modelled correctly, and FP is ‘false presences’ i.e. number of observed absences which are modelled as presences ([www.bccvl.org.au/support](http://www.bccvl.org.au/support)).

1) Accuracy is the proportion of correctly predicted cases:

$$Accuracy = (TP + TN) / \text{total}, \quad (2)$$

Where TP is ‘true presences’ and TN is ‘true negatives’ (the number of correctly modelled absences) ([www.bccvl.org.au/support](http://www.bccvl.org.au/support)).

## Appendix 3 Results of the Thorny Devil sensitivity tests



**Fig. A1.** Results of sensitivity testing of algorithms to alternative, i.e. non-default, configuration option settings, for the eight algorithms investigated in this study, for the Thorny Devil: (a) Maxent, (b) ANN, (c) GLM, (d) GAM, (e) MARS, (f) FDA, (g) SRE and (h) CTA. Selected results; only the results of sensitivity tests where algorithms showed more than minimal sensitivity to non-default configuration settings are shown. Values in brackets denote control settings.



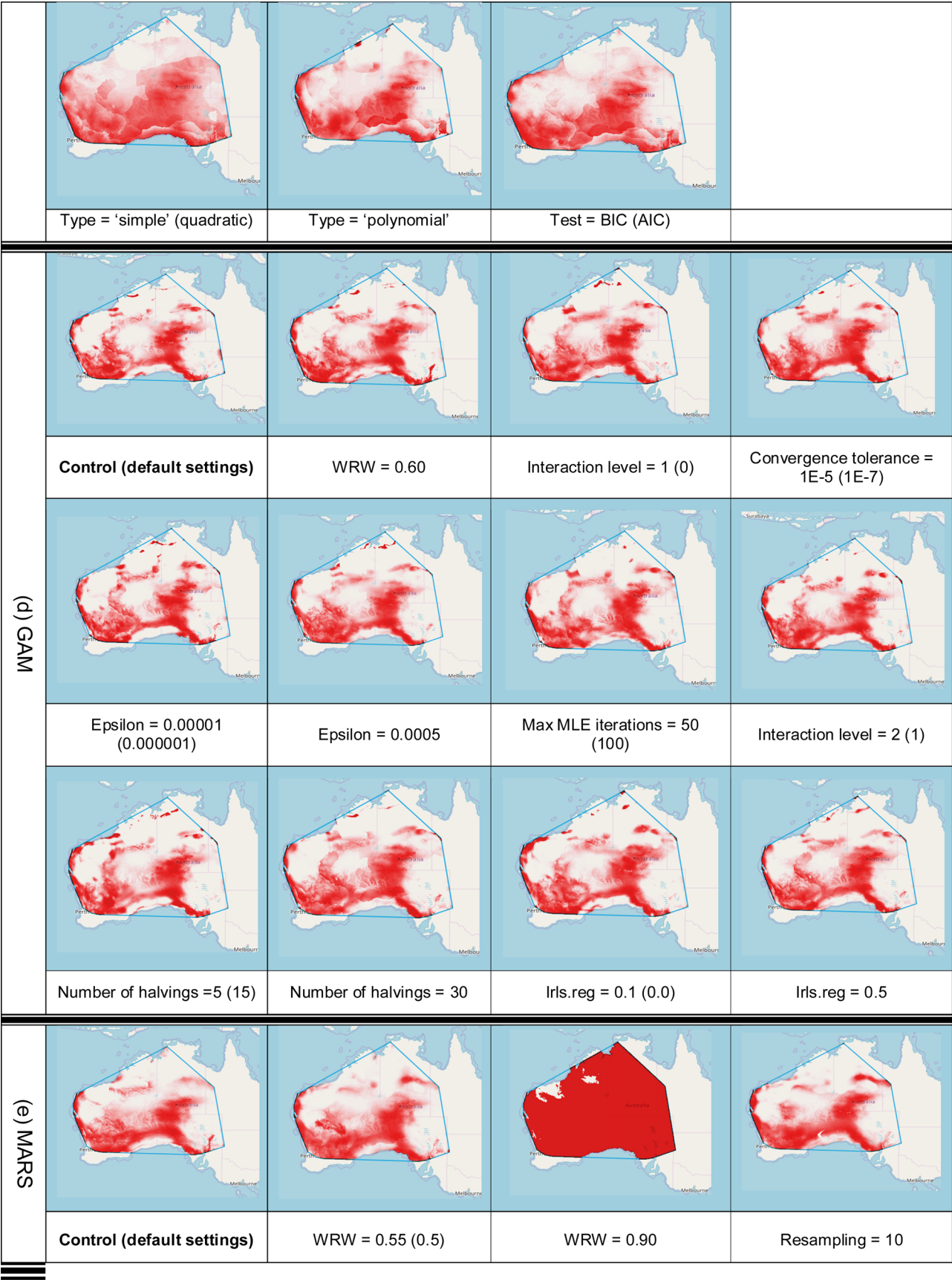


Fig. A1. (continued)

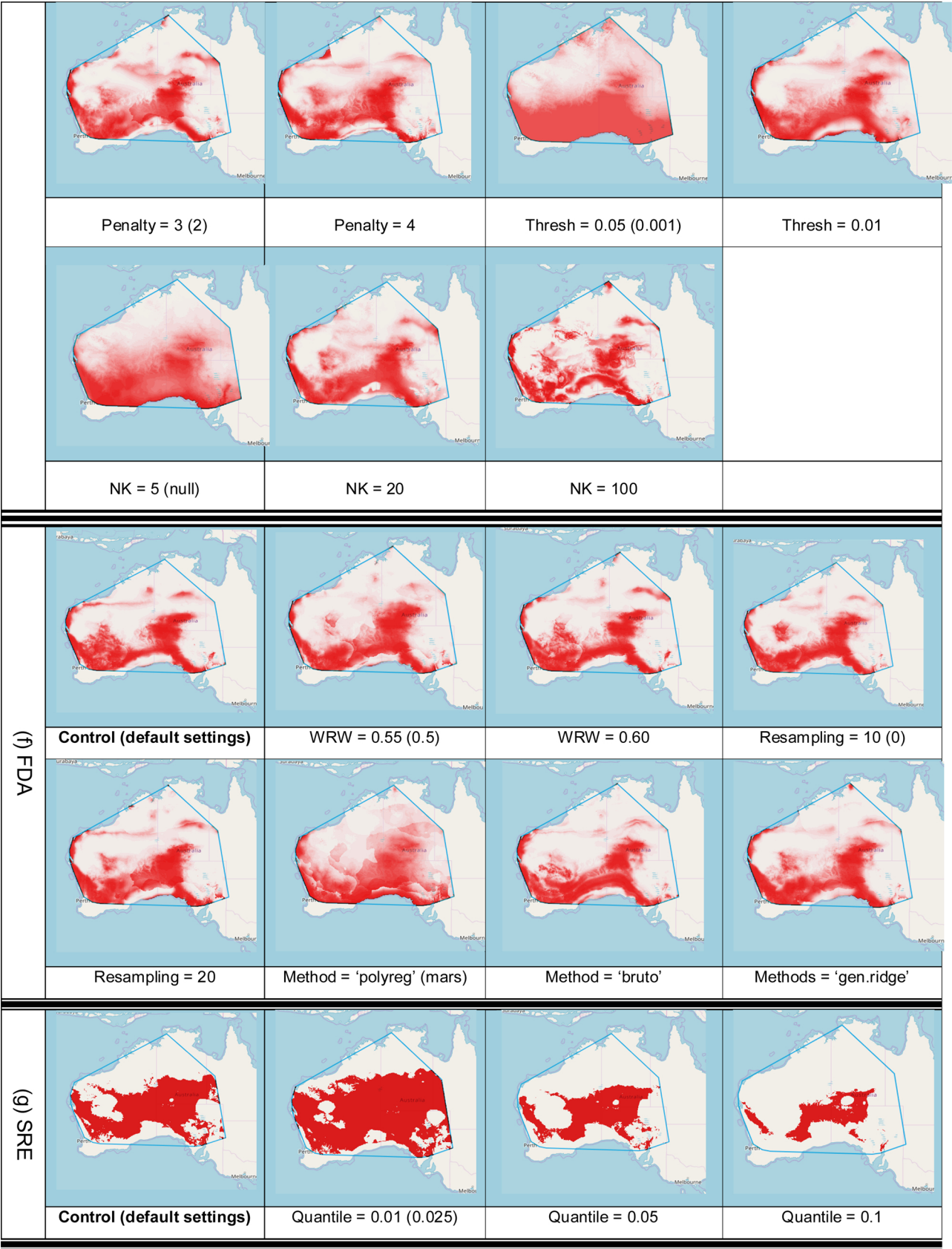


Fig. A1. (continued)

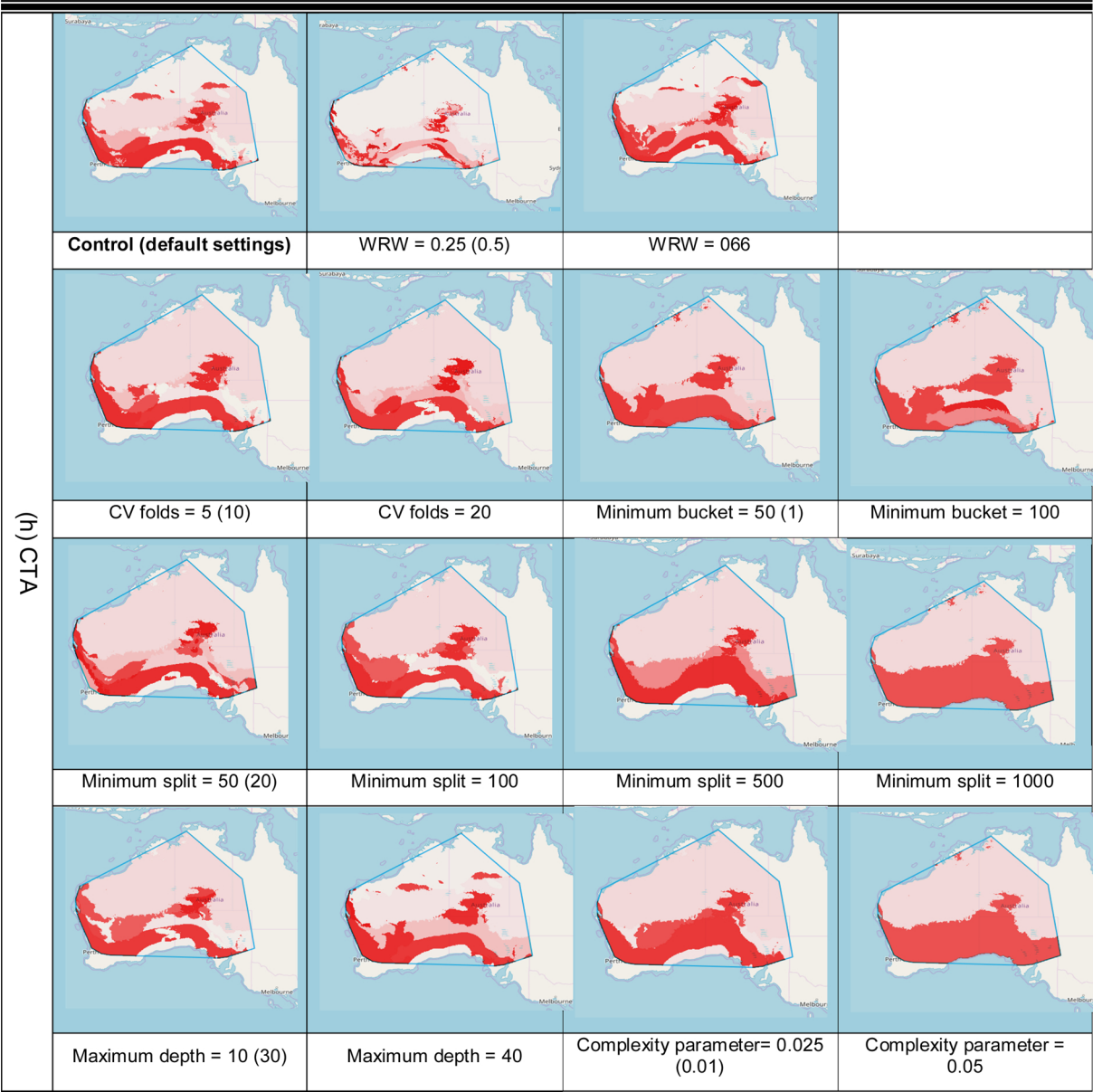


Fig. A1. (continued)

**Table A1**

Evaluation statistics used to diagnose algorithm sensitivity to configuration settings in the Thorny Devil sensitivity analysis: (i) Area Under the Receiver Operating Curve (AUC), (ii) Positive Predictive Value (PPV) and (iii) Accuracy, for each of the control (default settings) and test species distribution models (with alternative configuration settings). Only the results for test settings to which the algorithms showed sensitivity are shown here. Bolded numbers indicate larger sensitivity.

| Algorithm                        | Configurable options                            | Control AUC | Sensitive settings | Altered AUC | Control PPV | Sensitive settings | Altered PPV | Control Accuracy | Sensitive settings | Altered Accuracy |
|----------------------------------|-------------------------------------------------|-------------|--------------------|-------------|-------------|--------------------|-------------|------------------|--------------------|------------------|
| Maximum Entropy model (MaxEnt)   | Weighted response weights                       | 0.91        | –                  | –           | 0.24*       | 0.25               | 0.23        | 0.79             | 0.33               | <b>0.83</b>      |
|                                  |                                                 |             |                    |             |             | 0.33, 0.9          | 0.26        |                  | 0.66,              | <b>0.84</b>      |
|                                  |                                                 |             |                    |             |             | 0.66               | <b>0.27</b> |                  | 0.75               | 0.81             |
|                                  |                                                 |             |                    |             |             | 0.75               | 0.25        |                  | 0.9                | 0.82             |
|                                  | Maximum number of iterations                    | –           | –                  | –           | –           | 10                 | <b>0.28</b> | –                | 10                 | <b>0.84</b>      |
|                                  |                                                 |             |                    |             |             | 100                | <b>0.29</b> |                  | 100                | <b>0.85</b>      |
|                                  |                                                 |             |                    |             |             | 2                  | 0.21        |                  | 2                  | 0.77             |
|                                  |                                                 |             |                    |             |             | 5                  | 0.22        |                  | 5                  | 0.78             |
|                                  | Threshold feature regularization                | 2, 5, 10    | <b>0.88</b>        | <b>0.88</b> | –           | 10                 | 0.23        | –                | –                  | –                |
|                                  |                                                 |             |                    |             |             | 2                  | <b>0.28</b> |                  | 2                  | <b>0.84</b>      |
|                                  |                                                 |             |                    |             |             | 5                  | <b>0.33</b> |                  | 5                  | <b>0.87</b>      |
|                                  |                                                 |             |                    |             |             | 10                 | <b>0.30</b> |                  | 10                 | <b>0.85</b>      |
|                                  | Categorical feature regularization              | –           | –                  | –           | –           | 5                  | 0.23        | –                | 2                  | 0.80             |
|                                  |                                                 |             |                    |             |             | 10                 | <b>0.29</b> |                  | 10                 | <b>0.84</b>      |
|                                  |                                                 |             |                    |             |             | 2, 5               | <b>0.29</b> |                  | 2                  | <b>0.85</b>      |
|                                  |                                                 |             |                    |             |             | 10                 | 0.21        |                  | 5, 10              | <b>0.84</b>      |
|                                  | Linear/quadratic/product feature regularization | –           | –                  | –           | –           | ‘no’               | 0.21        | –                | ‘no’               | 0.76             |
|                                  |                                                 |             |                    |             |             | ‘no’               | 0.27        |                  | ‘no’               | 0.76             |
|                                  |                                                 |             |                    |             |             | ‘no’               | 0.22        |                  | ‘no’               | <b>0.83</b>      |
|                                  |                                                 |             |                    |             |             | ‘no’               | 0.22        |                  | ‘no’               | 0.78             |
|                                  | Hinge feature regularization                    | –           | –                  | –           | –           | PFT = 40           | <b>0.29</b> | –                | PFT = 40           | <b>0.85</b>      |
|                                  |                                                 |             |                    |             |             | QFT = 5            | –           |                  | QFT = 5            | –                |
|                                  |                                                 |             |                    |             |             | HFT = 8            | –           |                  | HFT = 8            | –                |
|                                  |                                                 |             |                    |             |             | –                  | –           |                  | –                  | –                |
|                                  | linear                                          | –           | –                  | –           | –           | 0.1                | 0.90        | –                | 0.25               | <b>0.85</b>      |
|                                  |                                                 |             |                    |             |             | 0.3                | <b>0.28</b> |                  | 0.25               | <b>0.80</b>      |
|                                  |                                                 |             |                    |             |             | 0.66               | <b>0.83</b> |                  | 0.3                | <b>0.82</b>      |
|                                  |                                                 |             |                    |             |             | 0.66               | <b>0.83</b> |                  | 0.66               | <b>0.83</b>      |
| Artificial Neural Network (ANN)  | Weighted response weights                       | 0.84        | 0.25, 0.33         | 0.87        | 0.79        | 0.25               | 0.80        | 0.88             | 0.25               | <b>0.80</b>      |
|                                  |                                                 |             |                    |             |             | 0.33               | <b>0.86</b> |                  | 0.3                | <b>0.82</b>      |
|                                  |                                                 |             |                    |             |             | 0.66               | <b>0.83</b> |                  | 0.66               | <b>0.83</b>      |
|                                  |                                                 |             |                    |             |             | 10                 | 0.78        |                  | 10                 | <b>0.80</b>      |
| Generalised Linear Model (GLM)   | NbCV                                            | 0.85        | 10                 | 0.86        | 0.84        | 20                 | <b>0.84</b> | 0.81             | 20                 | 0.85             |
|                                  |                                                 |             |                    |             |             | 0.25, 0.33,        | 0.84        |                  | 0.25               | 0.79             |
|                                  |                                                 |             |                    |             |             | 0.5                | 0.82        |                  | 0.33               | 0.79             |
|                                  |                                                 |             |                    |             |             | 0.66               | –           |                  | 0.5                | 0.80             |
| Generalized Additive Model (GAM) | Type                                            | 0.91        | simple             | 0.80        | 0.79        | polynomial         | 0.85        | 0.85             | 0.66               | <b>0.77</b>      |
|                                  |                                                 |             |                    |             |             | 1                  | 0.87        |                  | simple             | <b>0.76</b>      |
|                                  |                                                 |             |                    |             |             | 2                  | <b>0.88</b> |                  | polynomial         | 0.79             |
|                                  |                                                 |             |                    |             |             | –                  | –           |                  | 1, 2               | 0.79             |
| Generalized Additive Model (GAM) | Interaction level                               | 0.91        | None               | 0.83        | 0.79        | –                  | –           | 0.85             | BIC, None          | 0.78             |
|                                  |                                                 |             |                    |             |             | 0.55               | 0.9         |                  | –                  | –                |
|                                  |                                                 |             |                    |             |             | BIC                | <b>0.82</b> |                  | –                  | –                |
|                                  |                                                 |             |                    |             |             | 0.55, 0.60         | 0.80        |                  | 0.55               | 0.84             |
| Generalized Additive Model (GAM) | Test                                            | 0.91        | 0.55               | 0.9         | 0.79        | 0.55, 0.60         | 0.80        | 0.85             | 0.60               | 0.84             |
|                                  |                                                 |             |                    |             |             | 1, 2               | 0.80        |                  | 1                  | 0.84             |
|                                  |                                                 |             |                    |             |             | 0.1, 0.5           | 0.80        |                  | 2                  | 0.83             |
|                                  |                                                 |             |                    |             |             | 0.9                | <b>0.85</b> |                  | 0.1                | 0.83             |
| Generalized Additive Model (GAM) | Irls.reg                                        | 0.91        | 0.1, 0.5           | 0.90        | 0.79        | 0.1, 0.5           | 0.80        | 0.85             | 0.5                | 0.83             |
|                                  |                                                 |             |                    |             |             | 50                 | 0.81        |                  | 50                 | 0.83             |
|                                  |                                                 |             |                    |             |             | 150                | 0.92        |                  | 200                | 0.84             |
|                                  |                                                 |             |                    |             |             | 5                  | <b>0.86</b> |                  | 10                 | 0.84             |
| Generalized Additive Model (GAM) | Max MLE iterations                              | 0.91        | 50                 | 0.90        | 0.79        | 10                 | 0.80        | 0.85             | 20                 | 0.84             |
|                                  |                                                 |             |                    |             |             | 20                 | <b>0.83</b> |                  | 30                 | 0.83             |
|                                  |                                                 |             |                    |             |             | 30                 | 0.77        |                  | –                  | –                |
|                                  |                                                 |             |                    |             |             | 1E-5               | 0.81        |                  | –                  | –                |
| Generalized Additive Model (GAM) | No. of halvings                                 | 0.91        | 1E-5               | 0.91        | 0.79        | 0.000005,          | 0.81        | 0.85             | 0.000005,          | 0.82             |
|                                  |                                                 |             |                    |             |             | 0.00001            | 0.80        |                  | 0.00001,           | 0.84             |
|                                  |                                                 |             |                    |             |             | 0.00005,           | <b>0.83</b> |                  | 0.0001,            | 0.83             |
|                                  |                                                 |             |                    |             |             | 0.0005             | 0.89        |                  | 0.0005             | –                |
| Generalized Additive Model (GAM) | Convergence tolerance                           | 0.91        | 0.0001             | 0.91        | 0.79        | 0.0001             | 0.89        | 0.85             | 0.0001             | –                |
|                                  |                                                 |             |                    |             |             | 0.0005             | 0.89        |                  | 0.0005             | –                |
|                                  |                                                 |             |                    |             |             | 0.001              | 0.91        |                  | 0.001              | –                |
|                                  |                                                 |             |                    |             |             | 0.0005             | 0.89        |                  | 0.0005             | –                |

(continued on next page)



Table A1 (continued)

| Algorithm                                       | Configurable options      | Control AUC | Sensitive settings   | Altered AUC | Control PPV | Sensitive settings | Altered PPV | Control Accuracy | Sensitive settings     | Altered Accuracy |
|-------------------------------------------------|---------------------------|-------------|----------------------|-------------|-------------|--------------------|-------------|------------------|------------------------|------------------|
| Multivariate Adaptive Regression Splines (MARS) | WRW                       | 0.88        | 0.5, 0.55            | 0.86        | 0.82        | 0.55               | 0.79        | 0.82             | 0.5                    | 0.80             |
|                                                 |                           |             | 0.9, 0.99            | 0.52        |             | 0.60               | 0.77        |                  | 0.55                   | 0.80             |
|                                                 |                           |             |                      |             |             | 0.90, 0.99         | 0.51        |                  | 0.60                   | 0.81             |
|                                                 |                           |             |                      |             |             |                    |             |                  | 0.90, 0.99             | 0.52             |
|                                                 | Resampling                |             | 10                   | 0.89        |             | 20                 | 0.83        |                  | 100                    | 0.80             |
|                                                 |                           |             | 100                  | 0.87        |             | 100                | 0.80        |                  |                        |                  |
|                                                 | NK (max. no. of terms)    |             | 5                    | 0.81        |             | 5                  | 0.75        |                  | 5                      | 0.78             |
|                                                 |                           |             | 10, 15               | 0.84        |             | 10                 | 0.76        |                  | 10, 15, 20             | 0.79             |
|                                                 |                           |             | 20                   | 0.85        |             | 15                 | 0.75        |                  | 100                    | 0.84             |
|                                                 |                           |             | 100                  | 0.91        |             | 20                 | 0.77        |                  |                        |                  |
|                                                 | Penalty                   |             | 3                    | 0.86        |             | 3                  | 0.78        |                  | 3                      | 0.80             |
|                                                 |                           |             |                      |             |             | 4                  | 0.78        |                  |                        |                  |
|                                                 | Thresh                    |             | 0.05                 | 0.74        |             | 0.05               | 0.68        |                  | 0.05                   | 0.73             |
|                                                 |                           |             | 0.01                 | 0.85        |             | 0.01               | 0.78        |                  | 0.01                   | 0.81             |
|                                                 |                           |             | 0.005                | 0.87        |             | 0.005              | 0.81        |                  | 0.005                  | 0.81             |
| Flexible Discriminant Analysis (FDA)            | Weighted response weights | 0.89        | 0.5, 0.55            | 0.87        | 0.77        | 0.55               | 0.82        | 0.82             | 0.50                   | 0.81             |
|                                                 |                           |             | 0.6                  | 0.88        |             | 0.60               | 0.80        |                  | 0.55                   | 0.80             |
|                                                 |                           |             | 10, 100              | 0.87        |             | 10                 | 0.76        |                  | 10                     | 0.81             |
|                                                 |                           |             | 20                   | 0.88        |             | 100                | 0.80        |                  | 20                     | 0.81             |
|                                                 | Resampling                |             |                      |             |             |                    |             |                  | 100                    | 0.80             |
|                                                 |                           |             |                      |             |             |                    |             |                  |                        |                  |
|                                                 |                           |             |                      |             |             |                    |             |                  |                        |                  |
|                                                 |                           |             |                      |             |             |                    |             |                  |                        |                  |
|                                                 | Method                    |             | Polyreg              | 0.81        |             | Polyreg            | 0.76        |                  | Polyreg                | 0.76             |
|                                                 |                           |             | Bruto                | 0.87        |             | Bruto              | 0.78        |                  | Bruto                  | 0.81             |
|                                                 |                           |             | gen.ridge            |             |             | Gen.ridge          | 0.73        |                  | Gen.ridge              | 0.80             |
|                                                 |                           |             |                      |             |             |                    |             |                  |                        |                  |
| Surface Range Envelope (SRE)                    | WRW                       | 0.83        | 0.55, 0.6, 0.9, 0.99 | 0.84        | 0.99        | 0.55, 0.60, 0.99   | 0.93        | 0.85             | 0.55, 0.60, 0.90, 0.99 | 0.84             |
|                                                 |                           |             |                      |             |             |                    |             |                  |                        |                  |
|                                                 | Resampling                |             | 10, 20, 100          | 0.84        |             | –                  |             |                  | –                      | –                |
|                                                 |                           |             |                      |             |             |                    |             |                  |                        |                  |
|                                                 | Quantile                  |             | 0.01                 | 0.79        |             | 0.01               | 0.98        |                  | 0.01, 0.1              | 0.61             |
|                                                 |                           |             | 0.05                 | 0.74        |             | 0.05               | 1.00        |                  | 0.05                   | 0.74             |
|                                                 |                           |             | 0.1                  | 0.61        |             | 0.1                | 1.00        |                  |                        |                  |
| Classification Tree Analysis (CTA)              | Weighted response weights | 0.87        | 0.50                 | 0.82        | 0.86        | 0.25               | 0.85        | 0.84             | 0.25                   | 0.81             |
|                                                 |                           |             |                      |             |             | 0.66, 0.75, 0.90   | 0.82        |                  | 0.3, 0.66              | 0.83             |
|                                                 |                           |             |                      |             |             |                    |             |                  | 0.5                    | 0.82             |
|                                                 |                           |             |                      |             |             |                    |             |                  | 0.025                  | 0.79             |
|                                                 | Complexity parameter      |             | 0.025                | 0.80        |             | 0.025              | 0.80        |                  | 0.025                  | 0.79             |
|                                                 |                           |             | 0.05                 | 0.73        |             | 0.05               | 0.83        |                  | 0.05                   | 0.73             |
|                                                 | Cross validation folds    |             | 5                    | 0.84        |             | –                  |             |                  | 5, 20                  | 0.82             |
|                                                 |                           |             | 20                   | 0.85        |             |                    |             |                  |                        |                  |
|                                                 | Minimum bucket            |             | 50                   | 0.78        |             | 1000               | 0.85        |                  | 50, 100                | 0.77             |
|                                                 |                           |             | 100                  | 0.80        |             |                    |             |                  |                        |                  |
|                                                 | Minimum split             |             | 50                   | 0.85        |             | –                  | –           |                  | 50                     | 0.81             |
|                                                 |                           |             | 100                  | 0.84        |             |                    |             |                  | 100, 500               | 0.82             |
|                                                 |                           |             | 500                  | 0.82        |             |                    |             |                  | 1000                   | 0.74             |
|                                                 |                           |             | 1000                 | 0.74        |             |                    |             |                  |                        |                  |
|                                                 | Maximum depth             |             | 10, 20               | 0.84        |             | –                  | –           |                  | 10                     | 0.81             |
|                                                 |                           |             |                      |             |             |                    |             |                  | 20, 40                 | 0.82             |

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